HARMONY TUTOR

The intelligent tonal-harmony checker and multi-voice writing

USER MANUAL

Version 1.7 — August 2026
macOS · Windows

Contents

1. Introduction

1.1 What is Harmony Tutor

Harmony Tutor is a SATB (Soprano, Alto, Tenor, Bass) music editor with automatic, real-time harmonic analysis. It is designed for harmony students and teachers, musicians and composers who want to check and improve their four-part writing exercises.

The application analyses each chord the moment it is entered, identifies voice-leading errors, recognises cadences and modulations, and provides a didactic explanation for every flag. It is currently the only SATB editor with real-time harmonic analysis available.

1.2 Who it is for

- **Conservatory and music-school students:** immediate checking of exercises, understanding of mistakes, self-study.
- **Harmony teachers:** a fast correction tool, teaching support, time saved on recurring errors.
- **Self-taught musicians:** learning guided by practice and feedback.
- **Composers and arrangers:** voice-leading checking and harmonic analysis in a classical context.

1.3 Core principles

- **Spelling-First:** analysis is based on note names (spelling), not only on MIDI numbers. This guarantees correct distinction of enharmonic intervals and reliable recognition of augmented-sixth chords (It6, Fr6, Ger6).
- **Educational correction:** every violation shows a rule code, an expandable description and a practical tip for fixing it.
- **International academic standard:** nomenclature follows the academic standard, supporting both Roman numerals (functional figuring) and modern chord symbols (Cmaj7, Dm, G7/F, etc.).

1.4 System requirements

Operating system	Minimum requirements
macOS	macOS 10.15 (Catalina) or later
Windows	Windows 10 (64-bit) or later

The application does not require an internet connection. Recommended resolution: 1280×800 pixels or higher.

2. Installation and first launch

2.1 macOS

1. Download the Harmony Tutor-mac.dmg file from the website.
2. Open the DMG and drag “Harmony Tutor” into the Applications folder.

2.2 Windows

1. Download Harmony Tutor.win.exe from the website and run the installer.
2. If the “Windows protected your PC” (SmartScreen) warning appears, click “More info” and then “Run anyway”.
3. Follow the wizard. The app will be available in the Start menu.

2.3 First launch

On opening, Harmony Tutor presents an empty project in C major, 4/4 time, with 4 measures. The interface is in dark mode.

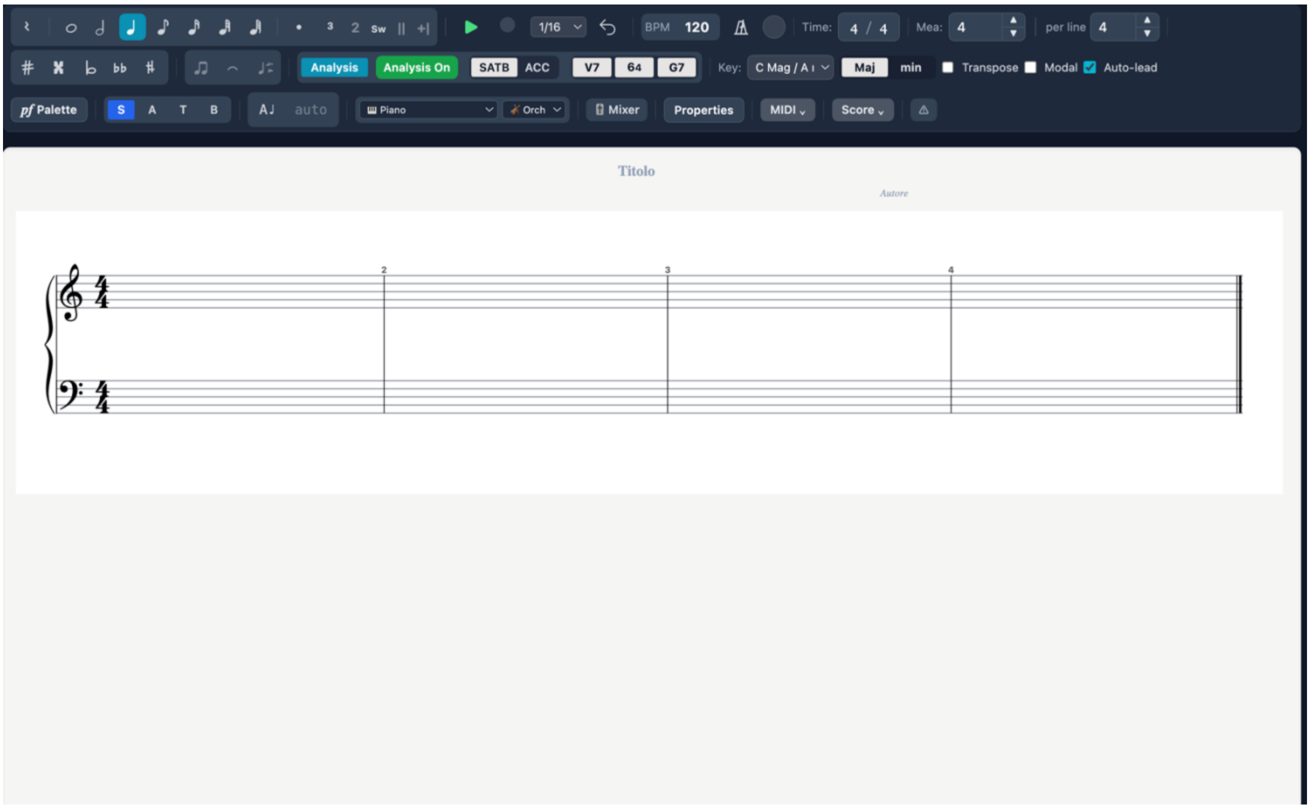


Figure 1 — First launch: empty project in C major, 4/4, 4 measures

2.4 New project (initial tracks and templates)

Creating a new project (File ‣ New, or the New action in the side menu) opens the New project dialog, where you set up the starting tracks before the project is created. You can always add or remove tracks later from the Mixer.

- SATB staff — the “SATB (4-voice choir)” checkbox shows or hides the four-part vocal staff. Uncheck it for an accompaniment- or percussion-only project.
- Accompaniment tracks — pick an instrument from the drop-down and click “+ Instrument” to add a pitched track; “+ 🥁 Drums” adds a percussion track. The chosen tracks are listed below; ✕ removes one.
- Templates — “Save current configuration…” stores the current set of tracks under a name; click a saved template to load it into the selection, 🗑 to delete it. Templates are kept locally and are offered for every new project.
- Create project / Cancel — confirm the configuration to create the project, or close the dialog without creating it.

3. The interface

3.1 Overview

- **Toolbar (top):** controls for voice, durations, accidentals, key, meter, playback and instrument.
- **Central editor:** the staff with the SATB voices, harmonic labels (Roman numerals) below the staff and chord symbols above (toggleable).
- **Analysis panel (right):** list of violations, severity filters, rule search, explanation of harmonic labels.

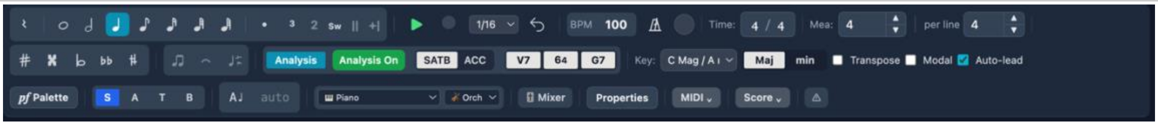


Figure 2 — The main toolbar

3.2 Display modes

The editor supports several layouts, selectable from the Preferences panel (Editor tab) or the side menu. Switching layout is instantaneous: the notes are unchanged.

Mode	Description	Staves
Grand Staff (default)	Treble clef + bass clef	2
Treble clef	A single staff	1
Old SATB clefs	Four staves with old clefs (S, A, T, B)	4
Close score	Grand Staff with reduced spacing	2
Open score	Separate staves with wide spacing	4

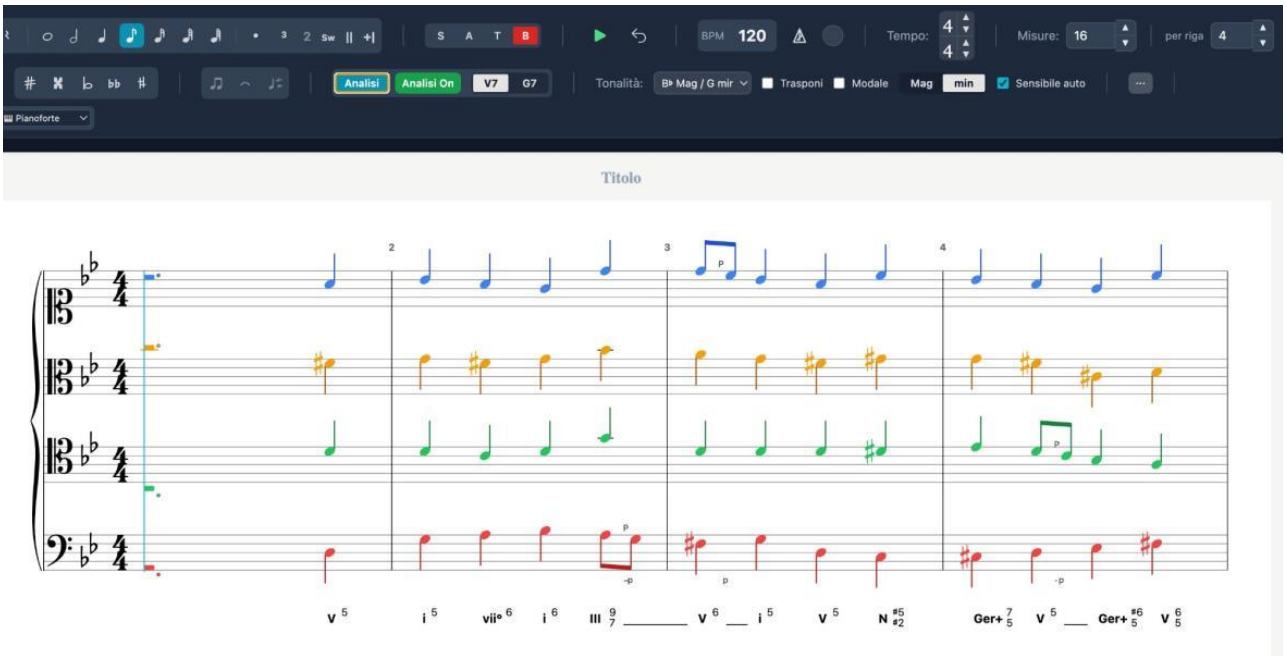


Figure 3 — SATB layout with old clefs, coloured voices, augmented sixths and Neapolitan

3.3 Side menu

The "..." button in the toolbar opens the side menu, which gathers the layout, format and MIDI controls.

- **Layout:** Grand Staff (2 staves), Treble clef (1 staff), Old SATB clefs (4 staves), Close score.
- **Format:** Page (portrait) or Landscape. Affects the on-screen layout and the PDF export.
- **MIDI:** enable/disable MIDI output and select the destination device (Internal Audio, IAC Driver Bus, Logic Pro Virtual In or other connected devices).
- **Step Input:** when active, notes are entered one at a time from the connected MIDI keyboard; the position advances automatically to the next beat.

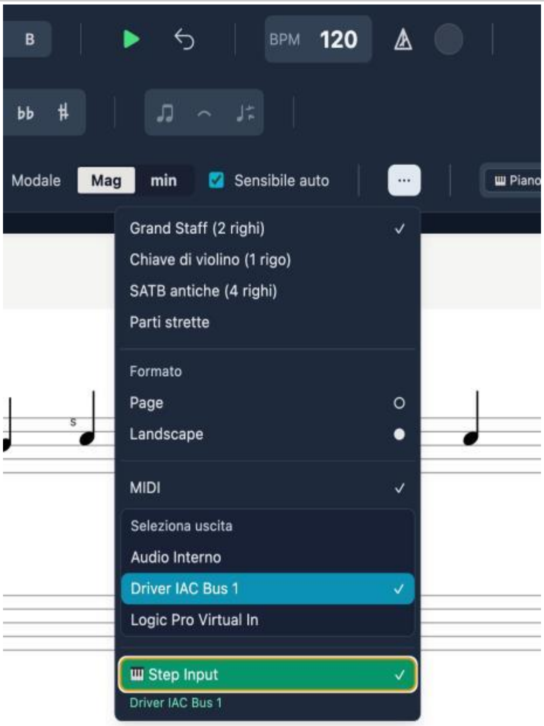


Figure 4 — Side menu: layout, format, MIDI and Step Input

3.4 Voice colours

By enabling “Voice colours (BTAS)” in Preferences → Editor, each voice is shown in a distinct colour: Soprano in blue, Alto in orange, Tenor in green, Bass in red.

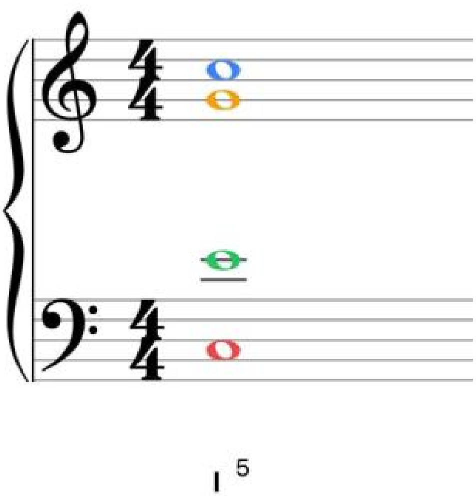


Figure 5 — Coloured voices: S (blue), A (orange), T (green), B (red)

3.5 Voice-leading indicators

On the staff, coloured dashed lines connect the notes between one chord and the next:

- **Green lines:** permitted or tolerated motion, or recognised cadences (PAC, IAC, HC, plagal). They appear only where the engine detects a meaningful pattern.
- **Orange lines:** a warning (situation to check).

- **Red lines:** a serious error (parallel fifths, parallel octaves, voice crossing, etc.).

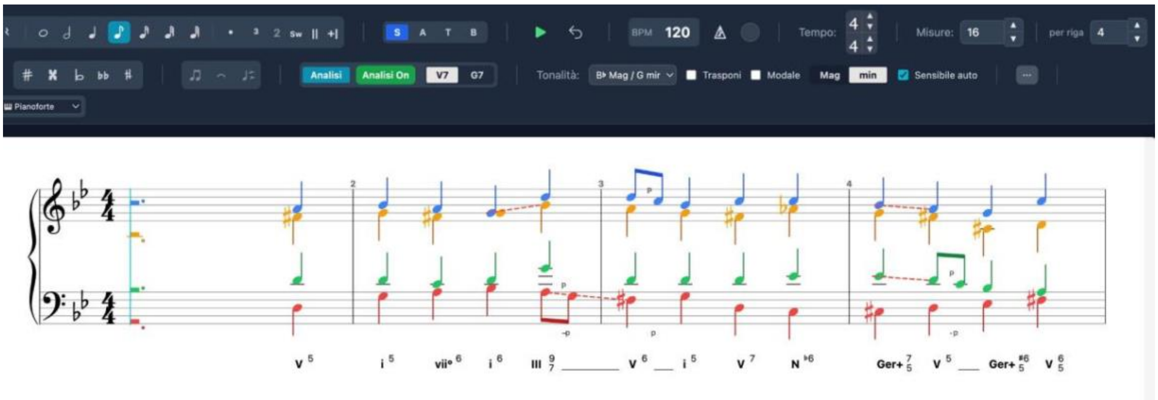


Figure 6 — Voice-leading with green, orange and red lines, Ger+ and Neapolitan

3.6 Chord symbols

In addition to Roman numerals, modern chord symbols (C, Am, D7/F#, Dm7/F, G7, etc.) can be shown above the staff. Toggle them from the toolbar selector or from Preferences → Analysis → "Show chord symbols".

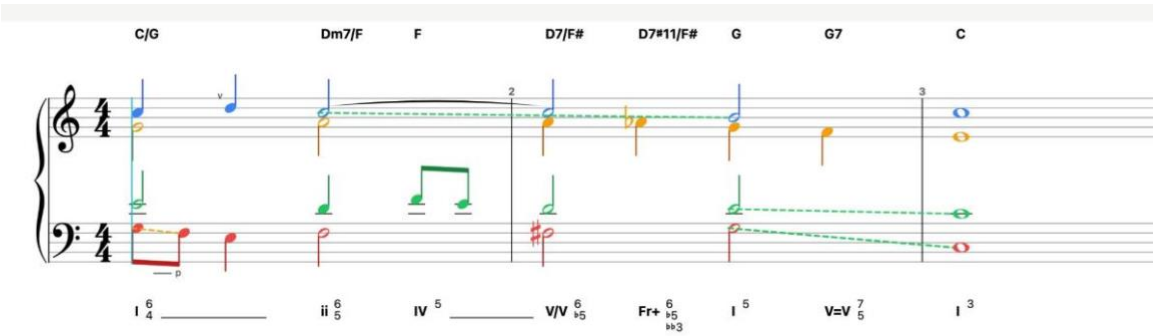


Figure 7 — Dual display: chord symbols above + Roman numerals below, with a French augmented sixth (Fr+)

3.7 Listening to single voices

To hear a single voice in isolation, double-click its voice button (S, A, T or B). Can be enabled on several voices. Double-click again to disable.

Alternatively, more appropriately, by going to the mixer panel accessible from the toolbar, you can solo, mute, or adjust the volume of any track.

More precisely: a double-click on a toolbar S/A/T/B button toggles SOLO for that voice (you can solo several at once). In the Mixer, every voice and ACC-track strip has its own M (mute) and S (solo) buttons, with unified semantics across voices and tracks: as soon as anything is soloed — a voice or a track — only the soloed channels sound, while mute always silences. This is the most flexible way to isolate or compare lines while writing.

3.8 Vocal ranges

Voice	Comfortable range	Extended range
Soprano	C4 – F5	B3 – A5
Alto	F3 – C5	E3 – D5
Tenor	A2 – E4	A2 – A4
Bass	E2 – B3	D2 – C4

3.9 Incomplete-measure warnings (⚠ toggle)

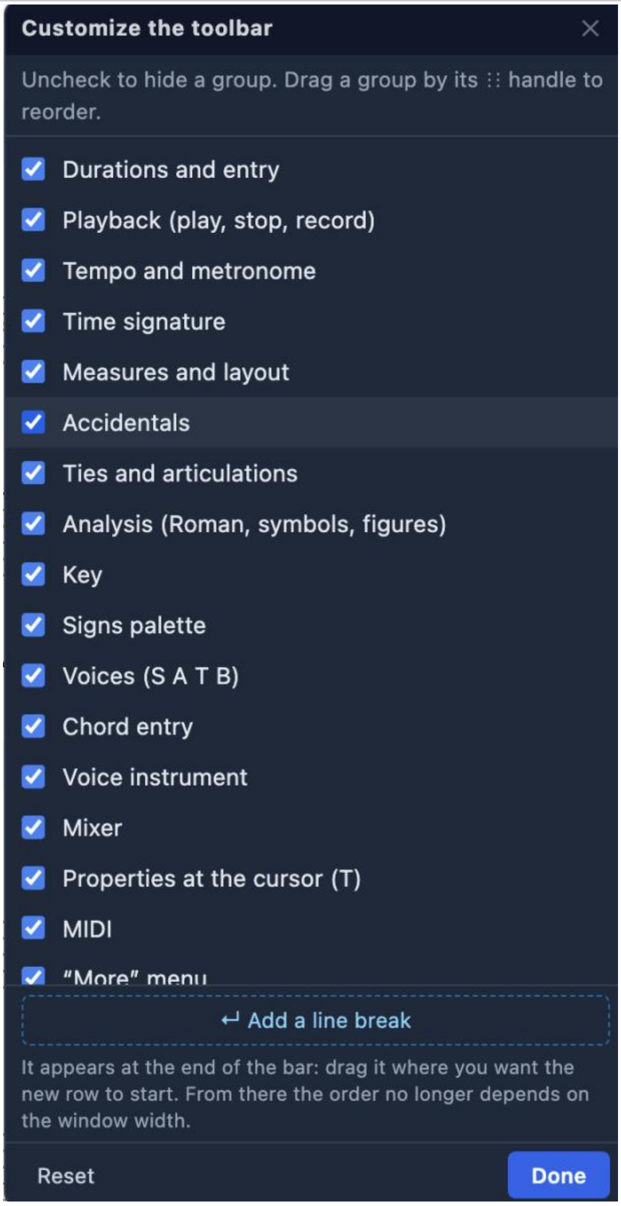
Measures that do not reach the duration required by the meter are flagged with a red rectangle. The warning can now be hidden:

- Click directly on the red overlay on the measure.
- **⚠ button in the toolbar:** red = warnings visible, grey = warnings hidden.

> ***Note:*** *The setting is not persistent: it resets to “visible” each time the file is opened.*

3.10 The command bar is yours to arrange

The toolbar is no longer the same for everyone. The customise command turns groups of buttons on and off: if you never use accompaniment tracks you can put them away, and what remains spreads out to fill the space.
Drag a group to change the order. The reordering happens when you release, not while you drag, so the row does not dance under your hand; and every point of the bar is a valid destination, the far end included.
A group can be sent to a NEW LINE, and it stays there: that is how you keep two short rows instead of one long row running off the screen.
When analysis is off, the commands that belong to it do not disappear: they stay where they are, greyed out. A button that leaves moves all the others, and your hand has to learn the layout again.



Customise the toolbar: groups are switched on and off, dragged to reorder, and a line break is added by hand.

4. Entering notes

4.1 Voice and duration selection

Action	Shortcut	Effect
Cycle voice	V	B → T → A → S → B
Whole note	1	4 beats
Half note	2	2 beats
Quarter note	3	1 beat
Eighth note	4	1/2 beat
Sixteenth note	5	1/4 beat
Thirty-second note	6	1/8 beat
Sixty-fourth note	7	1/16 beat

When you write several notes at the same point, the program can work out on its own which voice each belongs to, by looking at how they sit on the staff: the highest goes to the upper voice of that staff, the lowest to the bottom one. As long as the arrangement is unclear, the voice chosen in the toolbar applies.

4.2 Accidentals

Shortcut	Accidental	Effect
#	Sharp (♯)	+1 semitone
b	Flat (♭)	−1 semitone
n	Natural (♮)	Cancel
## (double)	Double sharp	+2 semitones
bb (double)	Double flat	−2 semitones

4.3 Modifiers

Key	Effect
.	Augmentation dot (duration ×1.5)
R	Toggle between note and rest
L	Toggle the tie on the current note
Option/Alt + F	Fermata — see 4.3a

4.3a Fermata

Applies a suspension of metronomic time to a note or rest, lengthening its actual duration in playback (default 2×). The playhead visually freezes and the following events shift accordingly.

1. Select the note or rest.
2. Press Option+F (Mac) or Alt+F (Windows).
3. The fermata appears above the note on the staff.

> ***Note:** To remove the fermata: select the note and press Option/Alt+F again.*

4.4 Ghost note and entry

Entering notes with the mouse requires the **Ctrl** (Windows) or **Cmd** (Mac) modifier key, to prevent accidental clicks.

1. **Select the voice** you want (S, A, T or B) from the toolbar or with the V key.
2. **Position yourself on the correct staff** (treble clef for Soprano/Alto, bass clef for Tenor/Bass). If voice and staff are inconsistent, the system blocks entry and shows no note.
3. **Hold Ctrl/Cmd:** the semi-transparent “ghost note” appears as a guide.
4. **Click** on the staff to insert the note definitively.

Without holding Ctrl/Cmd, the mouse acts only as a pointer and selection tool.

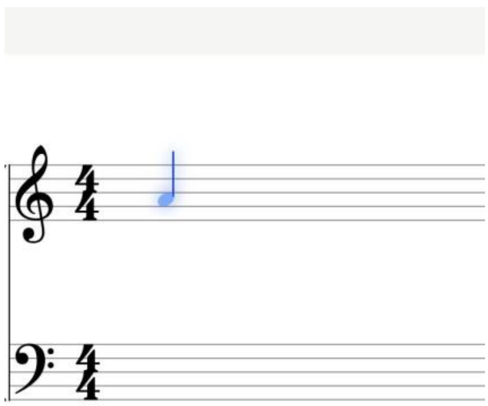


Figure 8 — Ghost note: preview before insertion

4.5 Selecting and editing

Action	Shortcut
Select note	Click
Multi-selection	Shift + Click
Delete	Backspace / Delete
Transpose ±1 semitone	↑ / ↓
Transpose ±1 octave	Alt + ↑ / ↓
Undo / Redo	Cmd/Ctrl + Z / Shift+Z

Deleting a note leaves a rest in its place: it keeps everything that follows from sliding backwards. What belonged to the note, though, goes with it — accidentals, fermatas, articulations, ornaments, stems and beams set by hand, and any ties or slurs involving it. Cmd/Ctrl+Z brings it all back.

4.6 Manual ornament marking

With the T panel open, the “Ornament marking” section lets you force the ornament type. Shortcuts use the Option (Mac) or Alt (Windows) key:

Shortcut	Type
Option+H	Structural (chord tone)
Option+P	Passing tone
Option+V	Neighbor tone
Option+A	Appoggiatura
Option+N	Anticipation
Option+S	Escape tone
Option+R	Suspension / Retardation
Option+O	Ornamental (figuration)
Option+C	Changing tone

Note: *On Mac the Control key is reserved for toggling note colours in the voices.*

What it applies to. The marking concerns the notes of the CHOIR: it tells the analysis whether a note belongs to the harmony or decorates it. On the notes of an accompaniment track the marking has no effect on the choir’s analysis, which reads the four voices: to make a track note count towards the choir’s harmony, use the collapse into a single chord (9.9).

4.7 Voice reassignment

Notes that have already been written can be moved from one voice to another by reusing the toolbar S/A/T/B buttons, without rewriting them. The feature works both in SATB editing and on “Grand staff with voices” ACC tracks.

1. Select the notes (rectangle selection is supported).
2. Click the destination voice (S, A, T or B) in the toolbar: the notes move to that voice.
3. The clef is recomputed from the voice: for example S → Tenor/Bass moves correctly to the lower staff.

Behaviour:

- **Collisions → chord:** if a note already exists at that position in the destination voice, the two become a chord (nothing is lost).

- **Automatic rests:** the source voice fills in rests where the moved notes used to be.
- **Undoable:** the operation is reverted with Cmd/Ctrl+Z.

> ***Note: the S/A/T/B buttons convert notes only when you made the selection yourself, by clicking a note or dragging a marquee. The note you have just written stays selected for convenience, but it is not converted: pressing another voice simply sets up the next note.***

4.8 Moving notes with the mouse

A note can be moved vertically by dragging it: press on the notehead and move up or down. The preview follows the mouse one scale degree at a time — line, space, line — and the pitch is written on release. If the note belongs to the current selection the whole selection moves, so a chord can be transposed in a single gesture; otherwise only that note moves. It works on accompaniment tracks too.

The move is diatonic: you change line or space and the spelling comes from the key signature (in F major, B comes out as B flat), exactly as when clicking to enter a note. The ↑↓ arrows remain chromatic, semitone by semitone: they are two different gestures. If any note of the group would fall outside the piano range, the move is not applied.

4.9 Enharmonic respelling (J)

The J key respells the selected notes with their enharmonic equivalent: D sharp becomes E flat and vice versa. The sound never changes; letter, accidental and staff position do. It is mostly needed on imported files, where a wrongly spelled note distorts the chord symbol, the figured bass and the scale degree (see 9.8, Inconsistent spelling).

4.10 Number of parts (4, 3 or 2)

The choir is not necessarily in four voices. Under More ‣ Parts you can choose 4 parts (S A T B), 3 parts (S A B) or 2 parts (S B). Disabled voices disappear from the toolbar and from the V key cycle, and can no longer be written.

The convention is that the outer voices always stay and the inner ones are removed: the lowest voice remains the Bass, so the rules that depend on it — inversions, doublings, motion of the outer parts — keep their meaning. Reducing the parts never deletes anything: if the voices to be switched off contain music, the change is refused with a warning. The number of parts belongs to the project and is saved in the file.

4.11 Where a note may start

Where a note lands does not depend on the duration you are about to write, but on what is already written. A measure offers three kinds of place: the beats of the metre (an empty measure is written from its beats), the END of every existing event — that is where what follows begins — and the START of every event, which is how you stack a chord on top of a note. The place nearest your click wins.
Rests count exactly like notes: to attack on an offbeat, write the rest that precedes it — which is also the correct notation. If you have written an eighth note, the next place is an eighth away, and a quarter note can start there.
Tuplets are a case of their own, because their attacks do not fall on the ordinary grid: a triplet of quarter notes does not sit on the beats. Inside a tuplet already begun the ordinary places do not apply; when only the last note is missing, the group can still be abandoned.
Hold SHIFT to fall back to the free grid, for attacks that do not grow out of what is already there.

How strongly a note is drawn towards existing attacks is set in Preferences → Editor ("magnet strength on attacks"): stacking a chord and stepping just off one are the same gesture read from opposite sides, and the right amount depends on how you work.

4.12 Voice-crossing warning

When the note you have just written ends up below (or above) its neighbouring voice, a warning appears. It is not modal: you keep writing, and the warning waits.

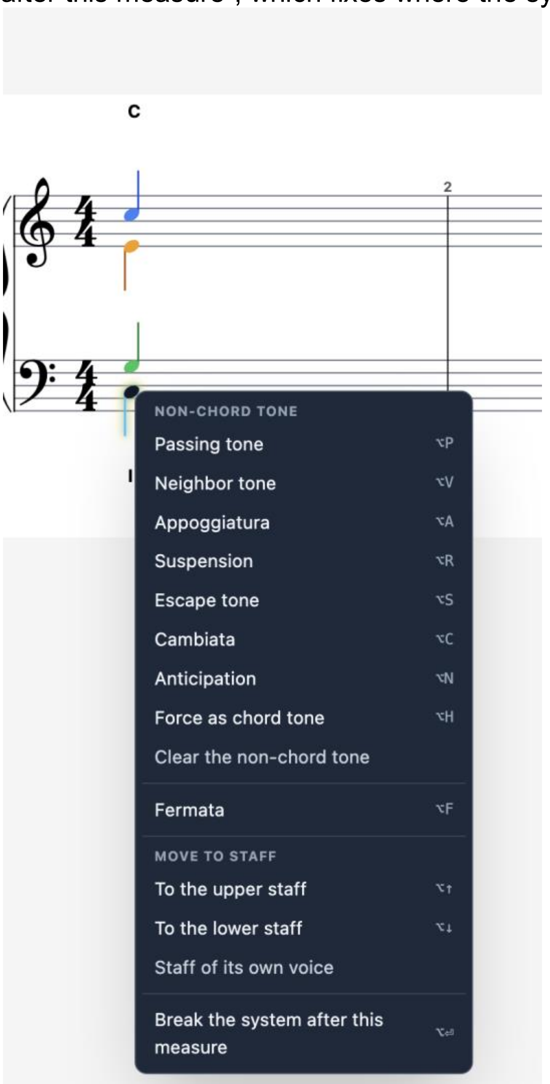
It offers three answers: swap the two voices with one click, accept it as it is, or turn these warnings off — anyone writing counterpoint with deliberate crossings does that straight away (Preferences → Editor turns them back on).

It exists because the analysis reports crossings anyway (rule R-04), but as a matter of PART WRITING, and only when you open the panel — possibly days later. When the real cause is a note entered in the wrong voice — which happens above all with WHOLE NOTES, the one figure with no stem to show whose it is — that report arrives in the right place at the wrong time, dressed up as a harmony problem.

4.13 The right-click menu on a note

Right-clicking a note opens a menu with everything that concerns that note: the non-chord tones (passing note, neighbour note, appoggiatura, suspension, escape note, cambiata, anticipation), "force structural" to declare it part of the chord, and "clear ornament".

Then the fermata, clearing articulations, moving the note to another staff (upper, lower, or the staff of its own voice) and "break the line after this measure", which fixes where the system ends.



The right-click menu on a note: non-chord tones, fermata, move to staff and system break.

5. Measure-edit panel (T)

Pressing T with the playhead on a measure opens the side panel with the editing tools.

5.1 Modulation / Tonicisation

Sets a key change from the current measure. Select the new tonic and mode (Maj/min), then “Apply context”.

5.2 Measure

- **Delete measure:** removes the current measure and shifts everything after it back.
- **Repeat:** inserts repeat barlines (open, close, double).
- **Meter change:** applies a meter change from the current measure.

5.3 Harmony override

Manually forces the harmonic label with three fields: Roman (e.g. I, V/vi, Ger+), Figure (e.g. 6/5) and Chord symbol (e.g. D7/F#).

5.4 Move to staff

Moves a note between the treble and bass staff with Option+arrow up/down (Mac) or Alt+arrow (Windows). Allows mixed open/close-score writing on the same system. “Restore default staff” returns the note to the staff assigned by its voice.

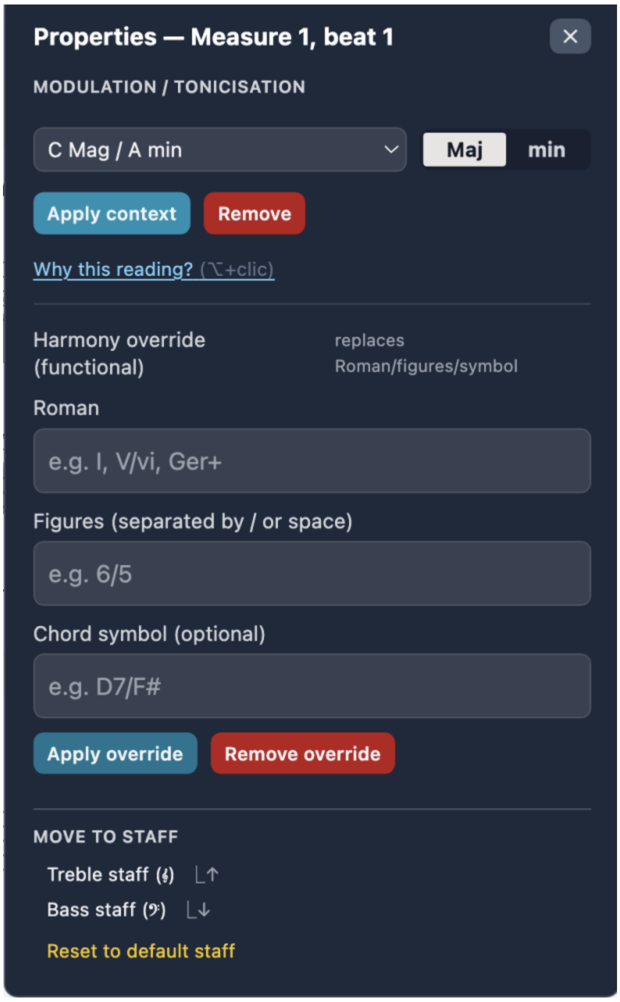


Figure 9 — The T panel: modulation, override, repeat, delete measure

6. Key and meter

6.1 Key selection

All 24 major and minor keys. Global change from the toolbar or per single measure from the T panel.

6.2 Minor mode

- **Natural (aeolian):** natural minor scale, with no raised leading tone.
- **Harmonic:** raised 7th degree to create the leading tone (V–i).

The “Auto leading tone” flag in the toolbar automatically inserts the leading-tone accidental in minor mode: the 7th degree is written already altered, without doing it on every note.

6.3 Modulations

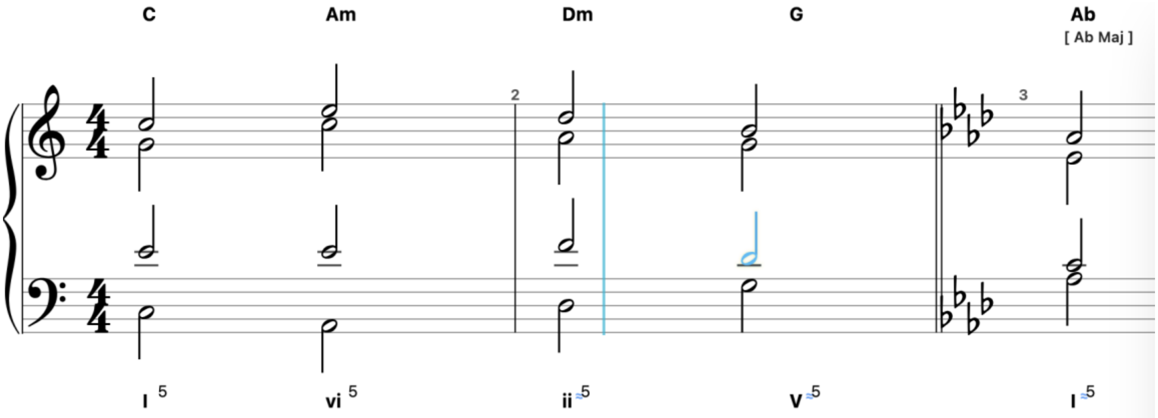
- **Automatic recognition:** cadential patterns tested against all candidate tonics.
- **Manual marking:** via the T panel.
- **Look-ahead tonicisations:** the engine looks up to 2 measures ahead.

6.4 Meter and BPM

Default meter: 4/4. Available: 2/4, 2/2, 3/4, 3/8, 4/4, 5/4, 6/8, 7/8, 12/8 and any custom meter. Meter changes from the T panel. Metronome toggled with K.

6.5 Key change within the piece

Besides the main key, the key signature can change at any point of the piece. In the signs palette, Structure group, pick the key and drag the ♯♭ button onto the measure it applies from: the new key signature appears, preceded by the naturals that cancel the previous one, and the following systems carry it at the start of the staff.



The key change at bar three: the double barline marks the boundary and the analysis carries on in the new key.

Notes are not moved. What changes is the key signature, and therefore which accidentals are printed: from there on you write in the new key, and clicking the A line in A flat major gives you an A flat, with nothing to add by hand.

It applies to the whole piece — choir and tracks alike — because the key belongs to the piece, not to the staff.

The analysis follows: from that point Roman numerals and figures read in the new key. The T panel is still there for when the two do not coincide, that is a modulation without a key change.

The change brings a double barline with it, which the analysis treats as a boundary: fifths and octaves across the change are no longer reported. Crossing a key change is not a voice-leading mistake.

To remove a change: right-click on the key signature, or «Remove key change» in the palette, which acts on the measure at the cursor.

7. Harmonic analysis

7.1 How it works

Harmony Tutor analyses the notes in real time through an 8-block pipeline. The result is a harmonic label (Roman numeral) below the staff for each chord and/or modern chord symbols above the staff, toggleable from the toolbar selector.

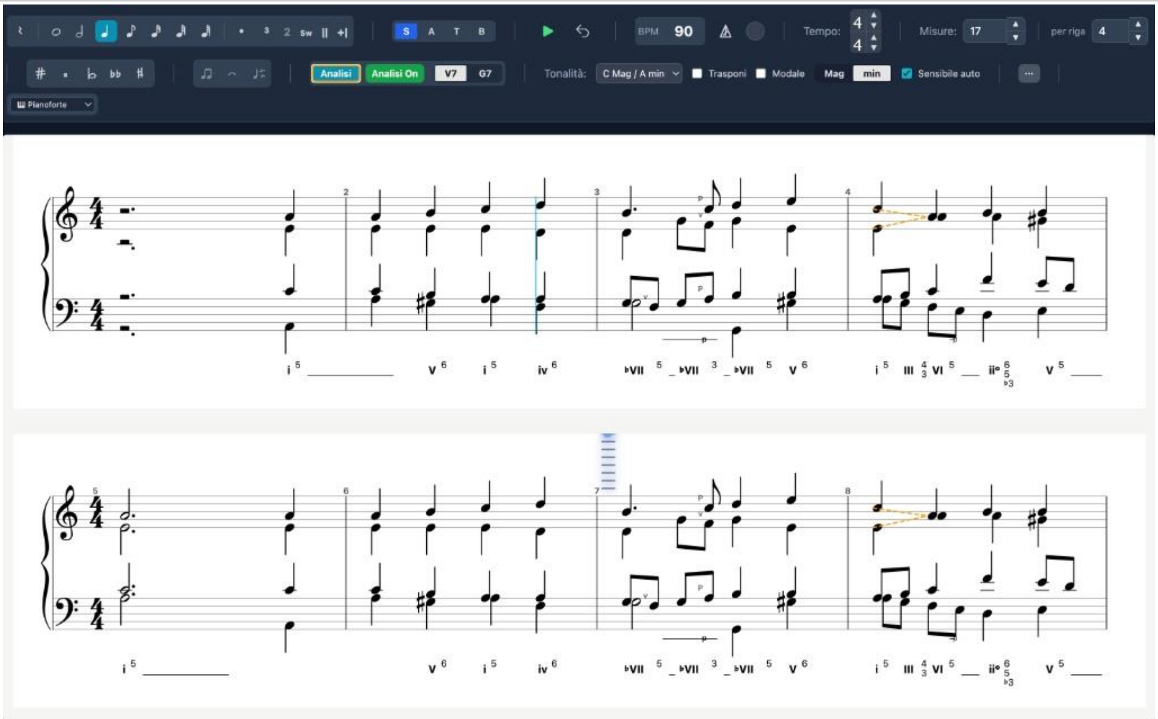


Figure 10 — Score with active harmonic analysis

7.2 The pipeline

The diagram below illustrates the full flow from note entry to label rendering.

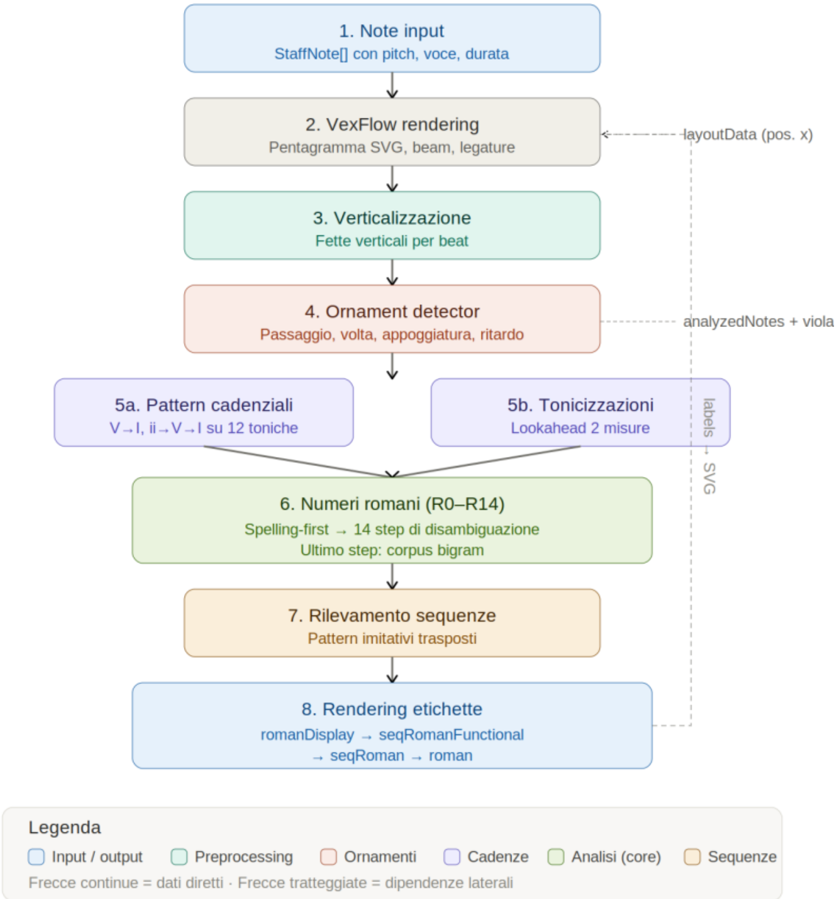


Figure 11 — Harmonic-analysis pipeline: 8 blocks

- **1. Verticalisation:** grouping of simultaneous notes.
- **2. Ornaments:** identification and separation of non-chord tones.
- **3. Chord identification:** spelling-first analysis, candidates ranked by plausibility.
- **4. Roman numeral (R0–R14):** 14 disambiguation steps; the last one uses the statistical corpus.
- **5. Cadential patterns:** recognition of cadences and tonicisations.
- **6. SATB rules:** over 40 voice-leading rules.
- **7. Sequences:** recurring melodic patterns (optional).
- **8. Rendering:** labels below the staff (and symbols above, if enabled).

7.3 Recognised chords

Triads

Type	Example (C)	Intervals
Major	C–E–G	1–3–5
Minor	C–E♭–G	1–♭3–5
Diminished	C–E♭–G♭	1–♭3–♭5
Augmented	C–E–G♯	1–3–♯5

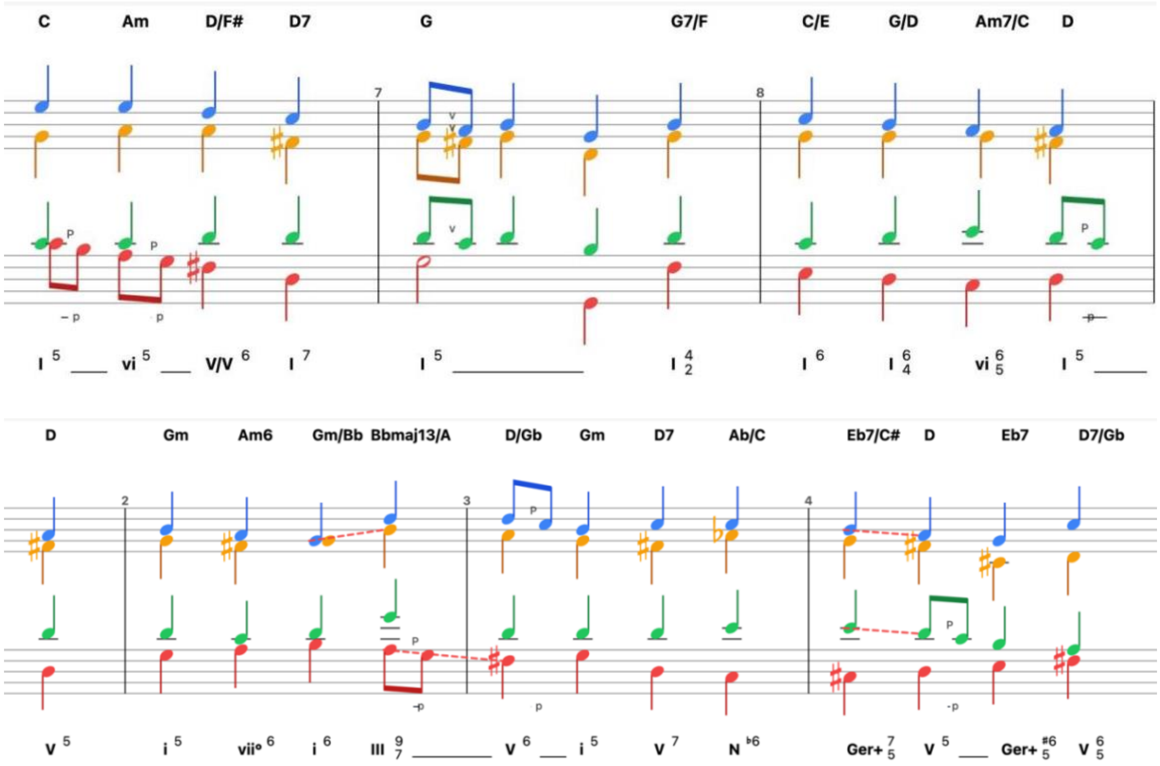
Sevenths

Type	Example (C)	Intervals
Dominant 7	C–E–G–B♭	1–3–5–♭7
Major 7	C–E–G–B	1–3–5–7
Minor 7	C–E♭–G–B♭	1–♭3–5–♭7

Type	Example (C)	Intervals
Diminished 7	C–E♭–G♭–B♭♭	1–♭3–♭5–♭♭7
Half-diminished	C–E♭–G♭–B♭	1–♭3–♭5–♭7

Special chords

- **Augmented sixths:** Italian (It6), French (Fr6), German (Ger6).
- **Neapolitan (♭II):** major triad on the lowered 2nd degree.
- **Dominant ninths:** V9, V7♭9, V7♯9.



Figures 12-13 — Coloured voices with symbols and Roman numerals, ornaments (P, v)

Analysis of instrumental tracks (ACC). The engine described in this chapter works on the SATB chorale. Accompaniment tracks have a dedicated harmonic analysis (chord symbol above the staff and Roman numeral below, with harmonic rhythm and manual overrides): see §11.8.

8. Non-chord tones

Harmony Tutor automatically recognises non-chord tones and excludes them from the chord.

Type	Marker	Feature
Passing tone	P	Stepwise between chord tones, weak beat
Neighbor tone	v	Steps away from and back to the real note
Appoggiatura	A	Dissonance on a strong beat, resolves by step
Anticipation	N	Anticipates a note of the next chord
Escape tone	S	Leap + stepwise resolution
Suspension/Retardation	R	Held note, resolves downward
Changing tone	C	Ornament with a leap from the dissonance

Suspensions are classified as 4–3, 7–6, 9–8, 2–1. Manual marking uses the Option (Mac) or Alt (Windows) key + the corresponding letter.

9. Rules and violations

The analysis panel flags violations with coloured badges. Every flag is expandable with a description and a tip.

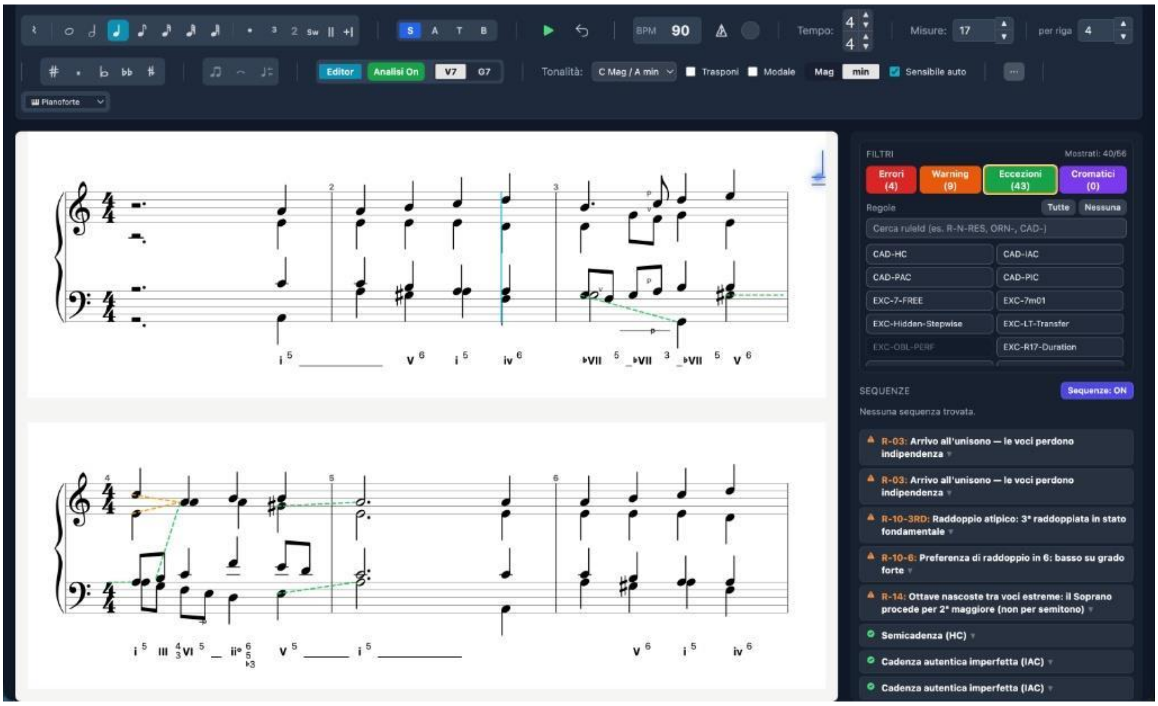


Figure 13 — Analysis panel with filters, violations and cadences

9.1 Severity levels

Colour	Severity	Meaning
Red	Error	Serious violation of fundamental rules
Orange	Warning	Situation to avoid, not always forbidden
Green	Exception	Special case tolerated by tradition
Purple	Chromatic	Advanced chromatic harmony

9.2 Errors (red)

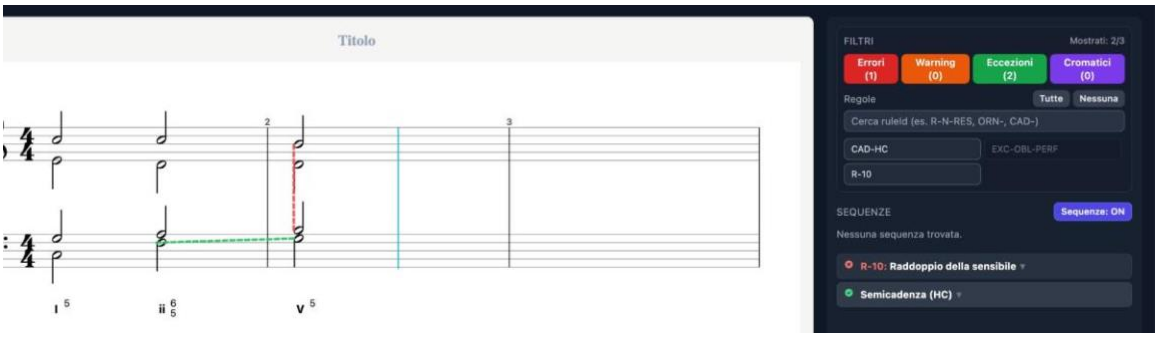


Figure 14 — Example error R-10 (doubled leading tone)

Code	Rule
R-01	Parallel octaves/unisons
R-02	Parallel fifths
R-04	Voice crossing
R-07	Leading tone does not resolve
R-09	Chromatic false relation
R-10	Doubled leading tone

Code	Rule
R-12	Seventh does not resolve

9.3 Warnings (orange)

Code	Rule
R-05	Hidden fifths/octaves
R-08	Excessive spacing S–A / A–T
R-13	Parallel motion in all voices
R-15	Wide leaps in an inner voice
R-16	Harmonic syncopation
R-17a/b/c	Problematic melodic leaps
R-RANGE	Note out of range

9.4 Exceptions (green)

Special cases tolerated by tradition, flagged with a green badge in the analysis panel (e.g. hidden octaves/fifths between outer voices moving by step).

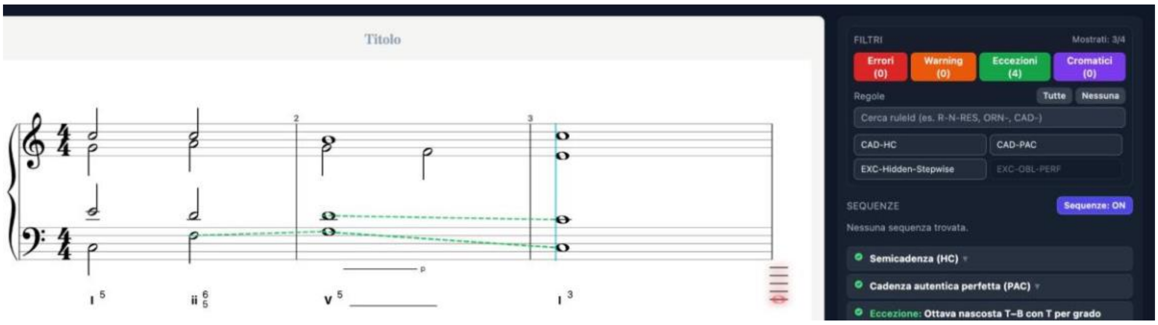


Figure 15 — Exceptions: PAC cadence, HC half cadence, hidden-by-step

Allowed syncopation: two exceptions to the barline rule

The “barline rule” (R-16) discourages a chord that enters on a weak beat and simply continues onto the downbeat of the next measure: the metrical accent would fall on harmony already heard, and the downbeat would carry nothing new. There are two cases where the writing is sound, and the analysis now recognises them as green exceptions.

The first is a change of INVERSION: if the chord returns on the downbeat over a different bass, the harmony is renewed and the accent finds a new event. It is the bass that decides this, not the upper parts.

The second is a RHYTHMIC PATTERN: if the same anticipation returns on the same beat for at least three consecutive measures, it is not an oversight but a deliberate figure — typical of ostinato accompaniment and compound metres. With only two measures the repetition may be accidental, and the rule stands.

Along with this, every rule that speaks of strong and weak beats now reads the metre IN FORCE in that measure rather than the piece's global metre: in a piece that moves between 3/4 and 4/4 the third beat changes nature, and the judgement used to be wrong wherever the metre changed.

9.5 Recognised cadences

Code	Type	Formula
CAD-PAC	Perfect authentic	V → I (root position)
CAD-IAC	Imperfect authentic	V → I (with inversion)
CAD-HC	Half cadence	Phrase ending on V
CAD-PLAG	Plagal	IV → I
CAD-DEC	**Deceptive cadence**	V → vi (maj.) / V → VI (min.) / V → ♭VI (chrom.)

Among the clues the analysis weighs when deciding whether a passage changes key or stays put there is also the half cadence: a pause on the dominant which, rather than announcing a modulation, confirms the key you are already in.

9.6 The analysis panel

- **Filters:** Errors, Warnings, Exceptions, Chromatic checkboxes with counts.
- **Rules:** clickable chips + search field (e.g. “R-01”, “CAD”).
- **Click a Roman numeral:** opens the Explanation card with all the details and alternative readings (≈).
- **Disable rule:** can be silenced individually; persists across sessions.

From the warning to the place where you fix it

Analysis labels are not only there to be read: clicking a roman numeral or a chord symbol takes you to where that reading is changed, without hunting for the right command in the panel.
Alternative readings — when a chord can be understood in more than one way — now appear where you write, next to the music, instead of sitting in a separate list.
A rule's advice can change with the situation, too. On a crossing between notes that are all STEMLESS, for instance, the advice no longer talks about part writing: it tells you to check which voice the flagged notes ended up in. On whole notes, a crossing is nearly always an entry mistake.

9.7 Cadential 6/4 → V$^{6}_{4}$

In the cadential progression I$^{6}_{4}$ → V → I, the I$^{6}_{4}$ is automatically relabelled as V$^{6}_{4}$, following the modern functional reading.

- Relabelling happens automatically when the engine detects the V$^{6}_{4}$ → V → I progression.
- Manual overrides (T panel) always take priority.

> ***Note:*** *With frequent modulations, check the tonal context before interpreting the V$^{6}_{4}$ label.*

9.8 Inconsistent spelling

When the notes as written do not form any chord — for instance D sharp, G and B flat, where D sharp to G is a diminished fourth rather than a third — the analysis reports Inconsistent spelling and points at the note to respell. The J key fixes it.

In that case the chord symbol shown is a reading of the sounds and does not match what is written: to make that explicit, it appears in parentheses. The Roman numeral survives the wrong spelling, because it is computed from the sounds; the figured bass does not, and shows distorted intervals until the note is corrected.

9.9 Collapse into a single chord (Opt+Shift+H)

This is for harmony spread over time rather than stacked: an arpeggio, for instance, produces confusing symbols note by note. Selecting the notes — even on different beats — and pressing Opt+Shift+H makes the analysis read them as one chord: the full label is pinned to the onset nearest the cursor. The other onsets lose their own label, so one chord no longer produces four identical symbols — except for those still sounding at that point, which are a harmony in their own right and keep theirs.

How to do it. 1) Move the cursor — the vertical line — to the point where you want the label to appear: it is the cursor, not the selection, that decides, because the command anchors the selected onset nearest to it. 2) Select the notes to be read together. 3) Press Opt+Shift+H. If the cursor is not placed, the label goes to the first onset of the selection.

With the analysis in SATB mode the command gathers the selected notes of the choir AND those of the accompaniment tracks. This is how a passage is disambiguated: select the choir notes together with the track note — typically a bass that clarifies an ambiguous chord — and the label is recomputed on the union. Selecting only the track notes, instead, derives the chord from those alone: it is the easiest mistake to make.

To undo it, repeat Opt+Shift+H on the same selection: the collapse is removed and the automatic analysis returns. Alternatively, use "remove override" in the T panel. Note that overrides are NOT undone by Cmd+Z, which acts on notes.

The label is not frozen. It remembers which notes it was made from and follows them: delete one of them and it is recomputed on the rest — take the B out of a C major seventh and it goes back to reading C — and when fewer than two are left it disappears, letting the automatic analysis return. Note that the lowest of the selected sounds decides the inversion, so annexing a note below the choir bass also changes the figure on the Roman numeral: the chord you declared has a different bass.

10. Audio and MIDI playback

10.1 Internal playback

Internal playback uses the Web Audio API. Instruments use good-quality sampled timbres (orchestral set, double bass and basses) loaded locally; timbres without local samples fall back to the General MIDI set (FluidR3). The default timbre is Piano. The full palette of selectable timbres (keyboards, strings, basses, woodwinds, brass and tuned percussion) is listed in 11.6 Available instruments.

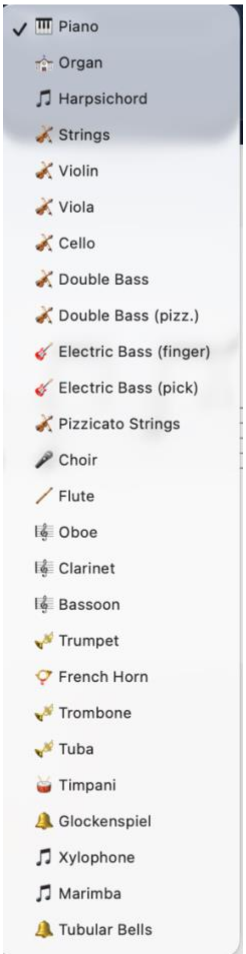


Figure 16 — Available timbres

Dynamics (velocity)

Each note now has a dynamics (velocity) value that affects playback and export. Velocity is captured from MIDI import and live recording, and is used for more expressive playback, for real-time monitoring and for external MIDI output. It is also preserved faithfully in MIDI export.

10.2 Controls

Shortcut	Action
Space	Play / Stop
Enter	Return to start
K	Metronome
+ / –	Tempo
Double-click S/A/T/B	Solo voice

10.3 External MIDI

Harmony Tutor supports MIDI keyboards on input and sending MIDI out to an external instrument or DAW (on Mac, via the IAC Driver). When an external MIDI output is selected:

- the internal piano (Web Audio) is silenced and the sound is monitored by the external instrument/DAW;
- control messages (CC), such as the sustain pedal (CC64), are forwarded, to avoid stuck notes;
- on stop, any notes still active are explicitly turned off (note-off flush).

External playback is velocity-sensitive. Configuration is done from the Preferences panel (MIDI tab) or the side menu.

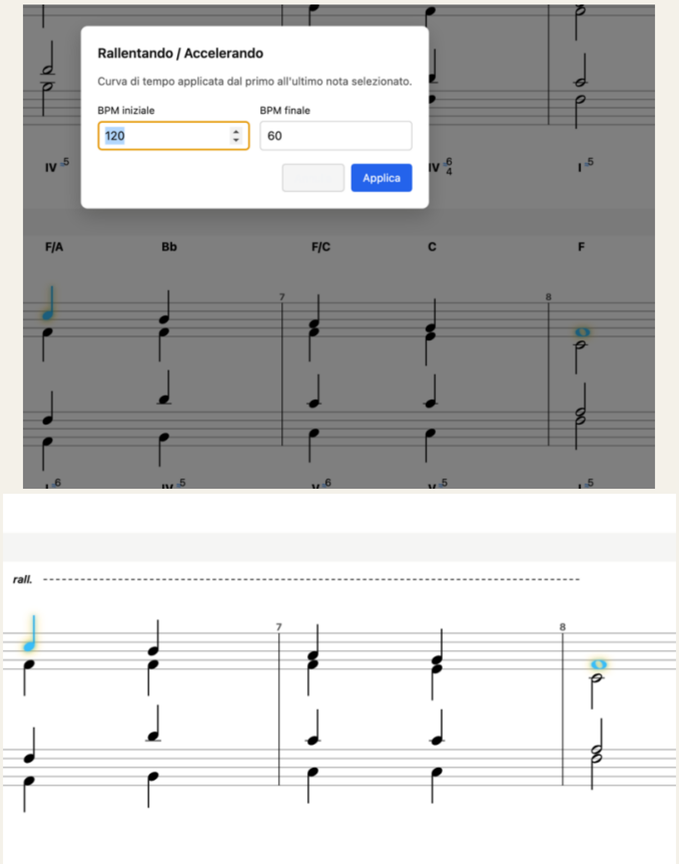
10.4 Tempo curve — Rallentando / Accelerando

The TempoCurve function creates gradual tempo changes over a range of notes. Playback interpolates the BPM linearly between the start and end value. The curve is shown as an overlay on the staff.

How to insert a tempo curve

1. Select the start and end note of the range (Shift+Click for multi-selection).
2. TempoCurve menu, or Shift+Option/Alt+R for a rallentando.
3. Set the start BPM and end BPM.
4. Confirm: the graphic overlay appears on the staff.

> ***Note:*** *The rallentando/accelerando also applies correctly to the notes of ACC tracks and to loaded files.*



TempoCurve — overlay on the staff with start and end BPM

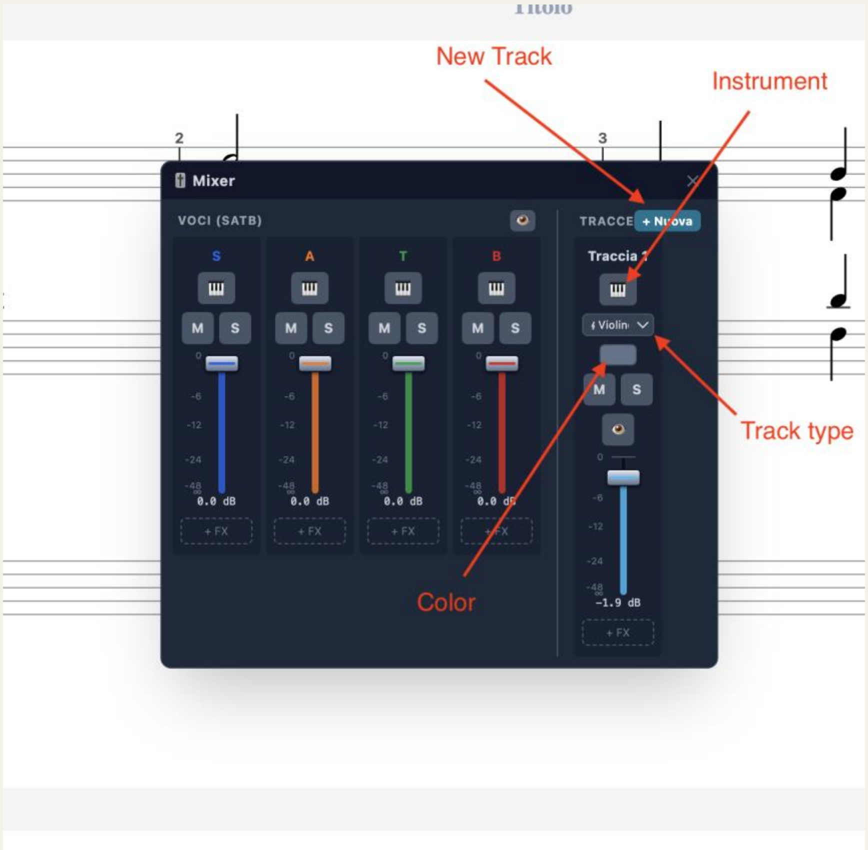
10.5 Restart the audio engine

If sound stops coming out — headphones unplugged, output device changed, audio system not ready when the app started — «Restart audio engine», in the Score menu of the toolbar, rebuilds the engine without closing the program. Your score is untouched.

It is needed because from the inside there is no way to tell that sound is not reaching the speakers: the system reports whether the engine is running, not whether sound comes out. A typical symptom is seeing the mixer meters move in silence.

11. Accompaniment tracks (ACC)

Accompaniment (ACC) tracks are independent of the SATB voices. They are ideal for figured bass, rhythmic patterns or arrangements, without interfering with the harmonic analysis of the chorale.



ACC tracks panel — add track, name, MIDI instrument

11.1 Adding and managing tracks

1. Open the tracks panel from the Mixer or the side menu.
2. Click the “+” button and choose the track type (see 11.4).
3. Assign a name and a MIDI instrument to the track.

ACC notes are visually distinct from SATB notes on the staff, including via the track colour. Right-clicking a Mixer strip deletes the track.

Orchestral section brackets

Two different brackets can appear to the left of the staves, answering two different questions. The black one gathers instruments of the same family — strings with strings, woodwinds with woodwinds — and is derived from the instrument chosen for the track. The blue one, further out, gathers the staves the analysis reads together. So an analysis group can span several families, and two instruments of the same family can be analysed separately.

11.2 Mixer

The Mixer is a floating, draggable panel that gathers control of the four SATB voices and all ACC tracks in a single window. It replaces the old sidebar mixer and opens from the Mixer button in the toolbar. The panel is split into two sections: VOICES (SATB) on the left and TRACKS on the right.

Mixer panel — VOICES (SATB) and TRACKS sections

Figure — Mixer: VOICES (SATB) and TRACKS sections

VOICES section (SATB)

One strip per voice: Soprano, Alto, Tenor, Bass. At the top of the section, a visibility button hides or shows the whole SATB block in the editor.

Control	Description
Instrument	MIDI instrument assigned to the voice.
M (Mute)	Mute / unmute the voice.
S (Solo)	Isolate the voice in playback.
Volume fader	Voice gain (0–100%), coloured with the voice colour. Adjustable without stopping playback.
MIDI channel	Outgoing MIDI channel for the external MIDI output: Auto (default) or a fixed 1–16.

TRACKS section (accompaniment)

One strip per ACC track. The “+” button (with its menu) adds a track; right-clicking a strip deletes it.

Control	Description
Track name	Editable label, shown in the track colour.
Staff / clef	Grand staff (treble+bass with a brace) or a single staff (treble, bass, soprano, alto, tenor). Bass and treble are also available as octave-down (8vb) transposing clefs (double/electric bass and guitar): they sound one octave lower than written, while the notation is unchanged.
Colour	Colour picker: vertical band to the left of the staff, tints the track's notes and the fader. SATB voices keep their fixed colours.
M (Mute)	Mute / unmute the track.
S (Solo)	Isolate the track in playback.
Visibility (eye)	Show / hide the track in the editor without removing it.
Volume fader	Track gain (0–100%).

Control	Description
MIDI channel	Outgoing MIDI channel for the external MIDI output: Auto (default) or a fixed 1–16 (Auto: voices 1–4, ACC tracks from 5 up, drums always 10).

Note: *Mute, Solo and Volume act in real time during playback (no need to stop and restart).*

Note: *The Mixer settings (instruments, volumes, mute, solo, colour and the track staff/clef configuration) are saved in the project file and restored on reopening. Older files without this data start from the defaults (piano, full volume, no mute).*

Adding tracks — the “+” menu

In the TRACKS header, a single “+” button opens a small menu: “New track” (a pitched accompaniment track) or “Drums” (a percussion track). This keeps the fader area uncluttered. The new track appears as a strip on the right.

Master faders (SATB, ACC and global)

Besides the per-channel faders, the Mixer has three master faders: SATB (controls the four voices together), ACC (controls all accompaniment tracks together) and MIX (the global output of the whole mixer). Routing is hierarchical — each channel feeds its group master, and both group masters feed the global master — so the SATB master scales only the voices, the ACC master only the tracks, and the MIX master everything. Each master has its own level meter, and the three values are saved with the project.

MIDI output channel

Every voice and track strip has a MIDI-channel selector for the external MIDI output (toward a DAW such as Logic): Auto (default) or a fixed channel 1–16. With Auto, the four voices use channels 1–4 and the accompaniment tracks are assigned from channel 5 upward; a drum track always goes to channel 10.

Changing the instrument of several voices/tracks at once

The instrument selector in the toolbar normally changes the active voice or track. If you first select notes belonging to several voices (or several ACC tracks), choosing an instrument applies it to all the voices/tracks present in the selection at once — handy for assigning, say, the same string sound to two inner voices in a single step.

Per-channel FX: EQ, Compressor, Pan and Reverb

Every strip — voice or track — has an insert effects chain: Pan (stereo position), a Reverb send, a three-band EQ and a Compressor. EQ and Compressor open in dedicated floating windows, plug-in style with graphs, from the EQ and Comp buttons at the bottom of the strip; a dot on the button shows the effect is active. In the EQ you can drag the three band points directly on the graph (the mouse wheel over the middle point adjusts its width, i.e. the Q); the Compressor shows the transfer curve and a gain-reduction meter. Reverb is a per-channel send: the type (Room, Hall, Plate) and the overall level are chosen on the master. The same EQ and Compressor are also available on the global master (MIX).

Group EQ and Compressor (SATB and ACC)

Besides the inserts on individual channels and on the global master, the two group buses — SATB and ACC — each have their own three-band EQ and Compressor, opened from the EQ/Comp buttons below their master faders. They act on the whole block: the SATB group EQ/Comp shapes all the voices together, the ACC group one all the tracks, before the signal reaches the global master. This is handy to 'glue' and balance the choir and the accompaniment separately. The settings are saved with the project.

Sound bank per voice and track: Orchestral or GM

Next to the instrument selector in the toolbar, a second menu chooses the sound bank of the active voice or track: Orchestral (the local orchestral samples, the default) or GM (the General MIDI soundfont). The two timbres have a different character — the orchestral one has a more realistic attack but can sound a little 'harder', GM is more uniform and softer — and they can be mixed freely, choosing a different one for each voice or track (for example GM on the inner parts and Orchestral on the melody). The choice is saved with the project.

Multiple channel selection and group actions

At the top of every strip there is a selection dot: click it to select the channel, and the strip is highlighted with a coloured border. Shift-click a second channel to also select every channel between the two. When at least one channel is selected, an action bar appears at the top of the mixer with the group actions: Mute and Solo (toggle all selected channels together), Delete (removes the selected ACC tracks — SATB voices cannot be deleted) and Deselect. In addition, dragging the fader of a selected channel moves all selected faders together by the same amount, preserving the group's relative balance.

Reordering and copying track settings

ACC tracks can be reordered from the right-click menu on the strip (Move left / Move right): the order changes both in the mixer and in the score. Also from the right-click menu, Copy settings and Paste settings transfer the clef and the whole effects chain (EQ, Compressor, Pan, Reverb) from one track to another, without touching the notes or the volume.
Mixer channels can be selected, and the Delete key removes the selected ones.
The equaliser window shows the spectrum of the sound, before and after the correction: you can see the effect of what you are adjusting instead of having to imagine it.

11.3 Accompaniment patterns

When the active staff is an ACC track, the toolbar shows the buttons for the chord-entry pattern. The chosen pattern determines how the inserted chord is written.

Button	Pattern	Description
Bl	Block	Full simultaneous chord.
Ar▲	Arpeggio up	Notes scaled upward.
Ar▼	Arpeggio down	Notes scaled downward.
Brk	Broken	Boom-chick: alternating bass and chord.
Alb	Alberti	Alberti bass pattern.
Ond	Wave	Up-and-down motion: ascending–descending.

Ped (Pedal / Let Ring) — unlike the previous patterns, this is not a way to insert the chord but an action applied to already-selected notes: it makes them ring until the next attack (pedal effect).

ACC pattern toolbar — Bl, Ar▲, Ar▼, Brk, Alb, Ond, Ped

Applying a pattern over a chosen voicing

Besides applying a pattern to a chord recognised by the analysis, you can select the exact notes you want — your own voicing — and apply the pattern to them. In this case the **exact pitches** selected are kept, with no automatic re-voicing: the pattern rhythmises and articulates precisely those notes.

The Arpeggiator panel

The decisions that shape an arpeggio — how the notes are laid out, the note value, the subdivision and the direction — are gathered in a single panel, instead of being spread between the toolbar and shortcuts.
The same pattern can be applied to several chords at once: select them and apply. And changing the note value does not use the chord up: the notes stay as they are, only the way they are spread over time changes.

11.4 Accompaniment track types

When creating a track with “+ New”, you can choose between several types, depending on the writing complexity needed:

Track type	Description
Single staff (Treble, Bass, Soprano, Alto, Tenor)	A single staff in the chosen clef. Suitable for melodic lines, basses, single-voice parts.
Grand staff	Double staff (treble + bass) with a brace, for two-staff writing.
Grand staff with voices	Double staff with the full voice management of the SATB (see below).

Grand staff with voices

This is the most articulated track type and the only one that allows true polyphonic piano writing. It reuses the same engraving engine as the SATB, applied to a single renamable track. It has four independent voices, distributed across the two staves:

Voice	Staff	Stems
Voice 1	Treble (right hand)	up
Voice 2	Treble (right hand)	down
Voice 3	Bass (left hand)	up
Voice 4	Bass (left hand)	down

Because the voices are independent, each can have its own rhythm (polyrhythm) and contain chords: this is the substantial difference from the simple Grand staff, which does not handle rhythmically independent voices.

> ***Note:*** *In practice, a “Grand staff with voices” track is equivalent to a SATB, but as a standalone, renamable accompaniment track: ideal for piano parts or multi-voice writing with rhythmic independence.*

On “voiced grand staff” tracks the voice a note will belong to is chosen by WHERE you click — as in the chorale — and the ghost note shows it in the colour of that voice before you write. A single voice can carry a chord: notes stacked on the same attack stay in the same part, under one stem.

11.5 Copy and paste between tracks

Copy/paste works between any tracks: between different ACC tracks and between SATB and ACC, in both directions. The flow is: copy the selection, choose the destination, paste.

- **Choosing the destination:** a single click on an ACC track's staff (or the SATB staff) sets that track as the active destination and positions the cursor, without inserting anything.
- **Automatic conversion:** pasted notes are adapted to the destination — on an ACC track they take the track's clef (or are split between treble and bass if it is a grand staff); in SATB they take the selected voice and clef.
- **Robust copy:** ACC-track selections and triplets are also copied correctly.

> ***Note:*** *Manual per-note markings (e.g. structural note, ornaments) are preserved during copy/paste between tracks.*

Copying an entire voice (button and shortcut)

To move a whole SATB voice onto an accompaniment track in one operation — instead of copying it line by line — copy the entire voice: right-click the S, A, T or B button in the toolbar and choose “Copy entire voice”, or select the voice and press Shift+Cmd/Ctrl+C. Then paste onto the destination ACC track as usual (Cmd/Ctrl+V, then click the track’s staff). This is the fastest way to send each of the four voices to a different instrument for a wider orchestration.

Usage tip — from chorale to orchestration

One of the most effective uses of copy/paste between tracks is to use the SATB as the harmonic foundation for an arrangement. The typical flow: write and correct the four-part chorale using the real-time analysis, until the harmony is valid and the part-writing is flawless; then copy the individual voices onto the accompaniment tracks, assigning each one an orchestral destination (for example the four voices to a string section). This way the parts enter the orchestral texture already correct in terms of harmony and voice-leading: the checking work done on the chorale is not lost, but becomes the solid base on which to build the orchestration.

Copying a whole track

Just as a whole choir voice can be copied, so can a whole accompaniment track: all its notes go to the clipboard and can be pasted wherever they are needed, even into another piece.

11.6 Available instruments

For every SATB voice and every accompaniment track you can choose the timbre from the instrument button in the Mixer. The sound palette has been expanded with sampled orchestral timbres, grouped by family:

- **Keyboards: Piano, Organ, Harpsichord.**
- **Strings: Strings (section), Violin, Viola, Cello, Double Bass, Pizzicato strings, Choir.**
- **Basses: Double Bass (pizzicato), Electric Bass (finger), Electric Bass (pick).**
- **Woodwinds: Flute, Oboe, Clarinet, Bassoon.**
- **Brass: Trumpet, Horn, Trombone, Tuba.**
- **Tuned percussion: Timpani, Glockenspiel, Xylophone, Marimba, Tubular bells.**

The chosen instrument applies both to internal playback and to external MIDI output. The basses (pizzicato double bass and electric basses) are natural-decay (one-shot) sounds, sampled across the instrument's typical low register; outside that range a note may not sound.

11.7 Drum track (percussion)

In addition to pitched accompaniment tracks, you can add a dedicated percussion track, on MIDI channel 10, to write drum-kit parts using standard notation.

Adding a drum track

- **In the Mixer, TRACKS section, press the button to add a percussion (drum) track.**
- **On creation the floating Percussion module opens automatically, from which you insert the hits.**

- **Alternatively, you can include a drum track right when creating a new project, from the initial-tracks dialog.**

Notation

The drum track uses standard percussion notation: a percussion clef and ✕ noteheads for cymbals. Each kit piece (kick, snare, hi-hat, toms, cymbals…) has its own fixed position on the staff.

The two kits

Two kits are available, selectable per track: Orchestral (default) and Rock. They change the sounds, the piece map and the overall timbre.

Entering hits

- **Percussion module: toggled with the 🥁 button in the toolbar (it appears only when a drum track exists). It shows the kit pieces; click a piece to insert it at the cursor position.**
- **With the mouse: click directly on the percussion staff at the desired point; the hit snaps automatically to the standard line for that piece.**

On playback the track plays the selected kit; on MIDI export it is sent on channel 10. If swing is active, the drum eighth notes also follow the triplet feel.

11.8 Harmonic analysis of ACC tracks

Besides the SATB chorale (Chapter 7), Harmony Tutor can harmonically analyse an instrumental piece written on an accompaniment (ACC) track: useful for studying the harmonic structure of an imported or freely composed piece. For each chord it shows the **chord symbol** above the staff and the **Roman numeral** below.
Activation. A global switch toggles the analysis subject between **SATB** and **ACC** (the two modes are mutually exclusive). In ACC mode you choose which track to analyse with the dedicated selector.
Harmonic rhythm. The engine automatically segments the flow into chords, detecting where the harmony changes even with irregular durations; a long note is taken into account in every beat where it sounds. Segmentation favours the points of greatest harmonic clarity.
Manual overrides. The automatic analysis can be corrected by hand:
• **Ornament marking** (Opt+O on a note): excludes that note from the chord reading — useful for passing or neighbour tones.
• Collapse to chord (Opt+Shift+H on a selection): declares that the selected notes form a single chord and writes the label where the cursor is. The same shortcut on the same selection undoes it. See §9.9.
ACC-track overrides are saved per track together with the project.

Combined analysis (parts on separate staves)

When a piece is spread across several ACC staves — typically the parts of a MIDI file imported as separate staves (see 13.1) — the analysis reads them as a whole. Tracks belonging to the same group are combined by onset and analysed together: the chord symbol and Roman numeral appear above the group's top staff and reflect the complete harmony, not the single part.
Analysis group. Multi-track import automatically assigns the same group to all parts imported together. To add or remove a track from the group, click the clef on its staff and use «Include in combined analysis» at the bottom of the menu. An unrelated track is not included until you add it explicitly; the group is saved with the project.

Setting the analysis to an ACC track. The group forms around the currently analysed ACC track, so first point the analysis at an accompaniment track: with analysis on, in the toolbar pick ACC in the SATB / ACC switch and choose the track to analyse from the drop-down that appears next to it. Then, from the clef menu of the other tracks, use «Include in combined analysis» to add them to the group.
Layout. Staves in the same group are joined on the left by a bracket and connected by the barlines, like a score system, and are shown closer together for an immediate overview.

Effects heard while auditioning

Volume, mute, solo, pan, EQ, compressor and reverb are heard when clicking or entering a note as well, not only during playback: auditioning goes through the same mixer channels. Adjusting an effect can be judged immediately, without starting playback.

Bringing back a hidden choir staff

Creating a project without SATB does not delete the choir staff but hides it: the four voices exist, empty. In the mixer, at the top of the "Voices (SATB)" column, the dot switches the staff on and off — the tooltip reads "SATB staff hidden — click to show". From there the choir comes back and can be written on.

11.9 Transposing instruments

Many wind instruments do not sound the note they read: a trumpet in B flat reading a C sounds a B flat. Harmony Tutor keeps the two apart — what sounds and what is written — and shows one or the other on request.
The file always holds the real sound. When a track is given a transposing instrument, its part is rewritten as the player reads it: the notes on that staff change, and so does that staff's key signature, while analysis, playback and export go on working from the true sound.
The «Concert pitch» checkbox switches between the two views: on, it shows what you hear; off, what each player reads on the stand. It affects the display only and moves nothing in the piece.

12. Real-time recording (REC)

Captures MIDI notes from the connected keyboard in real time, with an automatic count-in and post-recording quantisation.

REC button — active state (red) and count-in

12.1 Recording flow

1. Select the destination voice or ACC track.
2. Arm recording with **Shift+R** (requires a visible ACC track).
3. Press **Space** to start: the automatic count-in (default 1 measure) runs and recording begins at beat 1.
4. Play on the connected MIDI keyboard.
5. Press Stop (Space) to finish.
6. Post-recording quantisation is applied automatically.

> ***Note:*** *On ACC tracks the R key toggles note/rest; recording is armed with Shift+R.*

12.2 REC settings

Parameter	Description
Count-in	Number of count-in measures (default 1).
Quantisation grid	Minimum rhythmic value to which notes are snapped.
Overlapping-note trimming	Overlapping notes in the same voice are shortened automatically.

***Note:** Make sure the MIDI keyboard is configured in Preferences → MIDI.*

13. Import and export

13.1 Import

Format	Extension	Notes
Native	.htp / .json	Complete format
MIDI	.mid	See below
MusicXML	.xml	From MuseScore, Finale, Sibelius

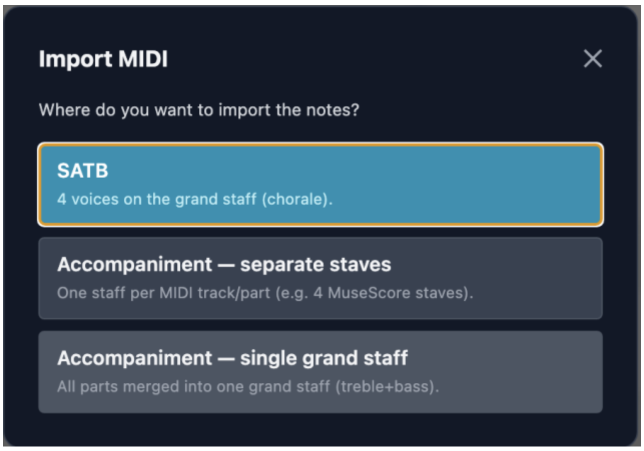
Advanced MIDI import

MIDI import is no longer limited to a simple heuristic voice assignment. When importing a MIDI file (e.g. a piano part) into an accompaniment track:

- **Automatic voice separation:** on a “Grand staff with voices” track, the part is rendered up to two voices per staff (fixed stems: voice 1 up, voice 2 down), without pre-splitting the tracks in a DAW.
- **Automatic clef assignment:** notes below middle C (MIDI 60) go to the bass staff, those from middle C upward to the treble staff.
- **Triplet-aware quantisation,** ties across the barline (a held note survives into the next measure) and correct measure counting, including for ACC-only imports.

Multi-track import and destination

When importing a MIDI file, a window asks where to place it, with three choices: SATB (four voices on the grand staff), Accompaniment — separate staves (one staff per MIDI track) and Accompaniment — single grand staff (all parts merged into two staves).
Separate staves. A multi-track MIDI file — for example a MuseScore score on four staves — is imported as one ACC track per part, each on a single staff, with the clef chosen from its range (treble or bass) and the name taken from the MIDI file. This is the right format for combined analysis (see 11.8); the single grand staff is still handy for a piano written on two staves.
Spelling in minor keys. MIDI does not store note spelling (C# and Db are the same key): with a minor key set, the leading tone is written with a sharp (e.g. C# in D minor) instead of the key-signature flat.



13.2 Export

Format	Extension	Notes
Native	.htp	Complete, reopenable
MusicXML	.xml	MuseScore, Finale, Sibelius
MIDI format 0	.mid	All voices on a single track
MIDI format 1	.mid	Each voice on a separate track
PDF	.pdf	For printing
PNG	.png	Screenshot

Export music

The File → Export music menu item (⇧⌘E) opens a single dialog with all export forms: MIDI (the piece's notes, for DAWs and other software), MusicXML (with the visible analysis — Roman numerals above and real figures below —, a universal format for MuseScore, Finale, Sibelius, Dorico…) and, in the «For blind users» section, the Spoken and Token modes described below. PDF and PNG export remain in their own menu items.

Printing, PDF and images

Printing and PDF fit the page to the sheet: if a system is wider or taller than the space available, everything is reduced just enough for each system to fit whole. On a score with many parts the reduction can be considerable — that is a matter of room on the paper, not a fault: fifteen staves on an A4 cannot come out any larger.
Image export produces several numbered files when the piece is long, and always cuts between one system and the next: no staff is ever split in half.
The paper format, portrait or landscape, is chosen in the layout settings. For a score with many parts portrait almost always works better, because height is the constraint.

What goes into the file: the three toggles and the staves

From version 1.6 the export follows what you see. The three analysis toggles — Roman numerals, chord symbols, figured bass — no longer govern the page alone: they also decide the contents of the files. What is off is not exported.
The same applies to the staves. A hidden staff — the choir or an accompaniment track — stays out of the file. This is how you export a single track, or two out of three: switch off what you do not need and export.
If no staff is left on, the export includes them all. A valid file with no music would be the worst outcome, because it looks like a successful export.
The commands live in the toolbar and in the View menu, with their shortcuts (§18.4). Before saving, the Export music dialog spells out what is about to be written — “Will be exported — Staves: Choir, Guitar. Analysis: Roman numerals, figured bass.” The summary is tied to the Export button, so a screen reader reads it together with the button.
MIDI is the exception for the analysis: it carries the notes, not the labels, so the three toggles do not affect it. The choice of staves does apply to it as well.

Chord symbols in the files

Chord symbols are exported as real chords, not as text: a <harmony> element in MusicXML and <Harmony> in MuseScore's native format. Whoever opens the file sees them as chords — they transpose together with the notes, they can be edited, and they are reprinted according to the conventions of the program reading them.
A symbol the application cannot classify is not lost: it is exported anyway, with its exact spelling. Better a chord declared “other” but spelled correctly than a missing symbol.

Accessible export of the analysis (MuseScore / VoiceOver)

Harmony Tutor can export the harmonic analysis in a form readable by a screen reader (VoiceOver), designed for blind students and musicians. From the **File → Export analysis…** menu (Cmd+Shift+X) a dialog opens with three modes:

• MusicXML — universal format (.musicxml): Roman numerals above the staff and figured-bass figures below. This is the format to open the file in any notation program (MuseScore, Finale, Sibelius, Dorico).
• **Spoken** — **MuseScore** format (.mscx): the analysis is written **in words** (e.g. “dominant seventh”) in the figured bass, in a small font. Navigating the bass notes in MuseScore, VoiceOver reads the phrase aloud. It requires no installation: the file is self-contained. (Verbalisation is currently in Italian.)
• **Token (for a VoiceOver dictionary)** — MuseScore format (.mscx) with compact labels in the figured bass: scale degree (V7, IV43) or actual chord symbol (Bb7, F#65). It requires a VoiceOver pronunciation dictionary that expands the labels into words.

About the formats. The **Spoken** and **Token** modes produce **.mscx** files, which only **MuseScore 4** can open — that is where the screen reader reads the analysis while navigating the notes. For a file that opens anywhere, use the **Standard** (MusicXML) mode.

Typical use (blind users). Export as **Spoken**, open the .mscx in MuseScore 4 with VoiceOver on and navigate the bass notes: for each chord you hear the scale degree with any inversion.

Chord symbols in words. In Spoken mode the symbols are said in full: Bb/D becomes “B flat bass D”, Gm(add9)/A “G minor with added ninth bass A”. The major mode is not named, because in a chord symbol it is implied: C is said “C”, Cm “C minor”.

Symbol and function together. If both Roman numerals and chord symbols are on, the file says both, separated by a comma: “B flat bass D, dominant seventh in first inversion”. The symbol says which chord it is, the Roman numeral says which function it has — and that is the comparison one makes while studying.

What the file contains (import dialog)

Before asking where to put the notes, the dialog lists what it found in the file: the name of each part, its number of notes, one or two staves and the clef. The "separate staves" option states how many staves will be created.

The instrument is taken from the file — from Program Change in MIDI, from the part list in MusicXML — instead of assigning a piano to everything. Percussion is recognised from channel 10 or from the track name and becomes a drum track, with the kit (orchestral or rock) chosen from the pieces the file actually uses. Next to each part a label states how it will be treated, "percussion" or "pitched": tap it to correct it. Percussion stays on its own track even when choosing "single grand staff", and with percussion only the SATB destination is not available.

Instrumental pieces. An imported piece (a guitar, a piano, the parts of a chorale) comes in as an accompaniment track, not as the choir. The accessible export follows it: it writes the notes the analysis is about, and if four parts are analysed together it exports all four.

Transposing clefs. The tenor of a chorale is written in treble clef with an 8 below: on import and export the program recognises and declares it, so the part stays where its author wrote it.

14. Automatic chorale generator

Generates the four SATB voices from a progression of Roman numerals. Chorale menu → “Insert progression”.

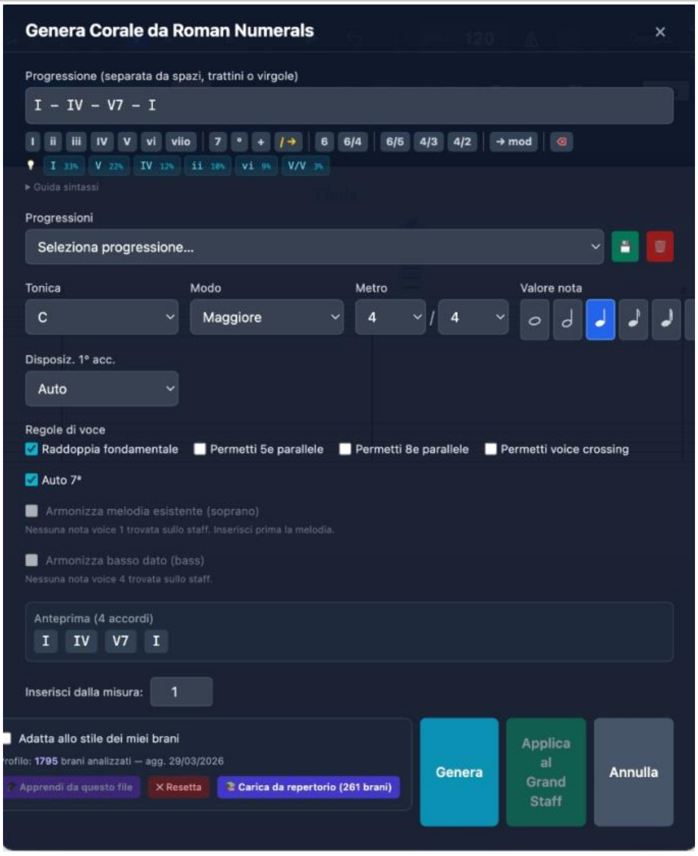


Figure 17 — Chorale generator

14.1 Interface

- **Progression field:** enter chords separated by spaces or dashes.
- **Quick buttons:** I, ii, iii, IV, V, vi, vii°, 7, °, +, 6, 6/4, 6/5, 4/3, 4/2, →mod.
- **Corpus suggestions:** coloured chips with the most likely progressions.

14.2 Settings

- **Tonic, Mode, Meter, Note value:** realisation settings.
- **Voice rules:** double the root, allow parallel 5ths/8ves, voice crossing, auto 7th.
- **Constraints:** given soprano melody or given bass as a fixed constraint.

14.3 Syntax

Syntax	Meaning
I ii IV V vi	Degrees (upper-case=Major, lower-case=minor)
I6 I6/4	Inversions
V7 V6/5 V4/3 V4/2	Seventh and inversions
vii° iiø7	Diminished / half-diminished
V/V V7/IV	Secondary dominants
It6 Fr6 Ger6	Augmented sixths
→G: →f#:	Modulation
|	Barline (new measure)

14.4 Corpus and learning

The generator uses a statistical system that learns from the pieces entered and analysed. Through bigram and trigram statistics, the software learns the horizontal behaviour of the voices and the most frequent harmonic progressions, guiding the choices of inversion, voice-leading and doubling.

- **Learn from this file:** adds the current exercise to the corpus.
- **Load from repertoire:** imports the pre-computed profile from the corpus.
- **Reset:** restores the default stylistic profile.

15. Teacher mode — Analysis Lock

Lets the teacher prepare exercises by locking the analysis engine and selectively hiding solutions or visual warnings. The lock is saved in the .htp project file: the student can open the file, view the score and play the audio, but the interface hides the elements chosen by the teacher. It cannot be bypassed without the password.

[SCREENSHOT TO BE INSERTED]
Analysis Lock modal — password field and masking options

Figure — Analysis Lock panel

15.1 Activating the lock

1. Help menu → “Analysis Lock”.
2. Set a teacher password.
3. Select the elements to hide.
4. Click “Apply lock”.

15.2 Maskable elements

Option	Effect when locked
Coloured lines and errors	Hides voice-leading flags and disables the violations panel.
Roman-numeral labels	Hides the Roman numerals below the staff.
Chord symbols	Removes the symbols above the staff.
Recognised ornaments	Hides the ornament markers (P, v, A, etc.).
Alternative readings	Removes the ≈ indicator.
Disable export and note copy	Blocks export (MIDI, MusicXML, PDF, PNG) and note copy/paste.

15.3 Unlocking

Click the padlock icon and enter the teacher password. To prevent accidental bypassing, there is no keyboard shortcut for this function.

> ***Note:*** *There is no password-recovery mechanism. Keep it in a safe place.*

16. About the application

Help menu → “About Harmony Tutor…” shows: the installed version, copyright and the website URL.

The PDF manual travels inside the application: Help menu → “User manual (PDF)…” opens it in the language of the interface. There is nothing to download separately, and it is always the one matching the installed version. In the same menu, “Shortcuts…” opens the full list of keyboard commands.

> ***Note:*** *On Windows this is the only way to check the installed version number.*

17. Preferences and settings

The Preferences panel is organised into six tabs. Settings are saved automatically.

17.1 Editor tab

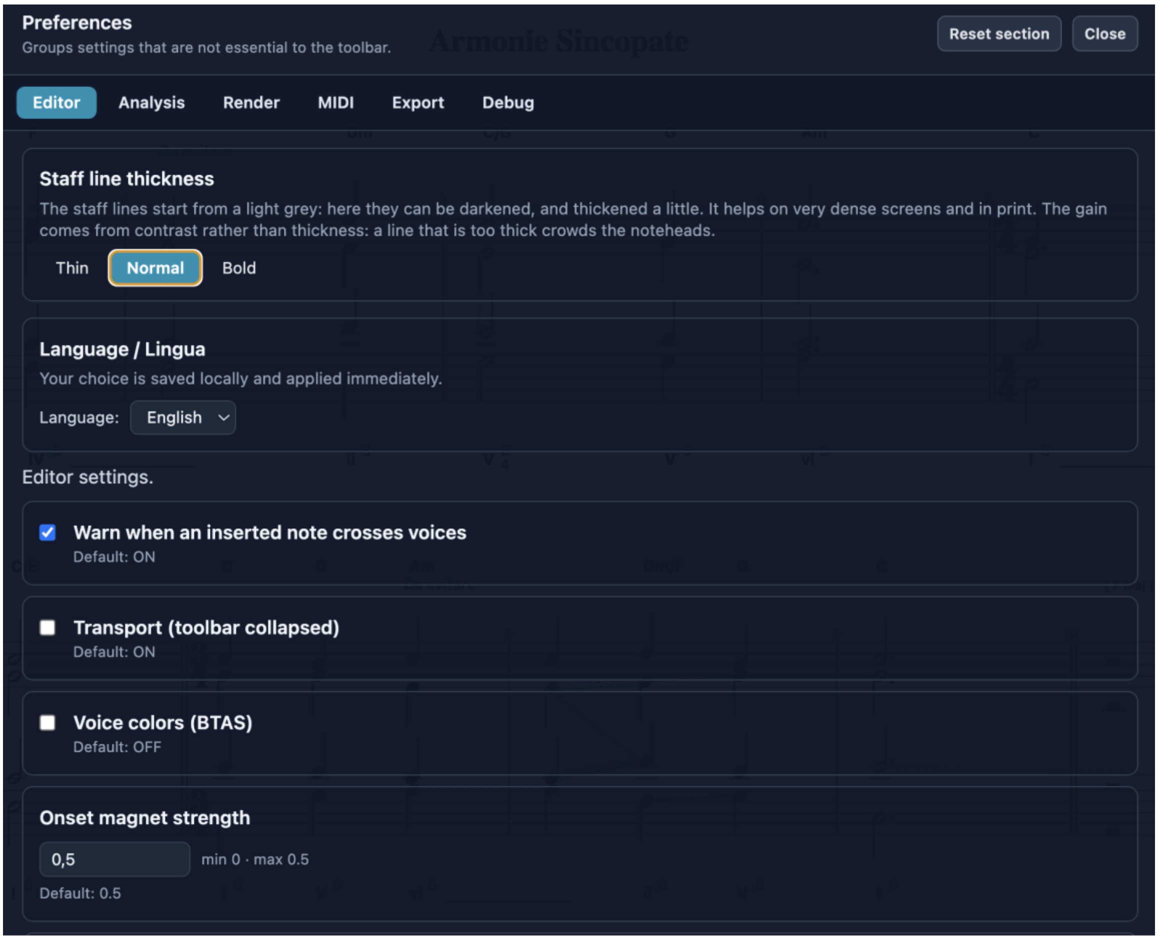


Figure 18 — Preferences: Editor tab

- **Voice colours (BTAS):** distinct colouring per voice.
- **Staff layout:** Grand Staff, Old SATB clefs, Treble-only, Close score, Open score.
- **Hide toolbar / Measure numbers / Autosave:** interface options.
- **Select current voice only:** rectangle selection filtered by voice.
- **Transport:** keeps the transport controls visible with the toolbar hidden.

Magnet strength on attacks — how strongly the note you are placing is drawn towards attacks already written in that measure (0 to 0.5 of the reference value). At zero only the grid applies; raising it makes stacking a chord easier and stepping slightly off one harder.
Warn when an entered note crosses the voices — shows the warning described in 4.12, with the option to swap the two voices. Anyone writing counterpoint with deliberate crossings turns it off.
Staff line thickness — the staff lines can be given more body, for printing or for very dense screens.

17.2 Analysis tab

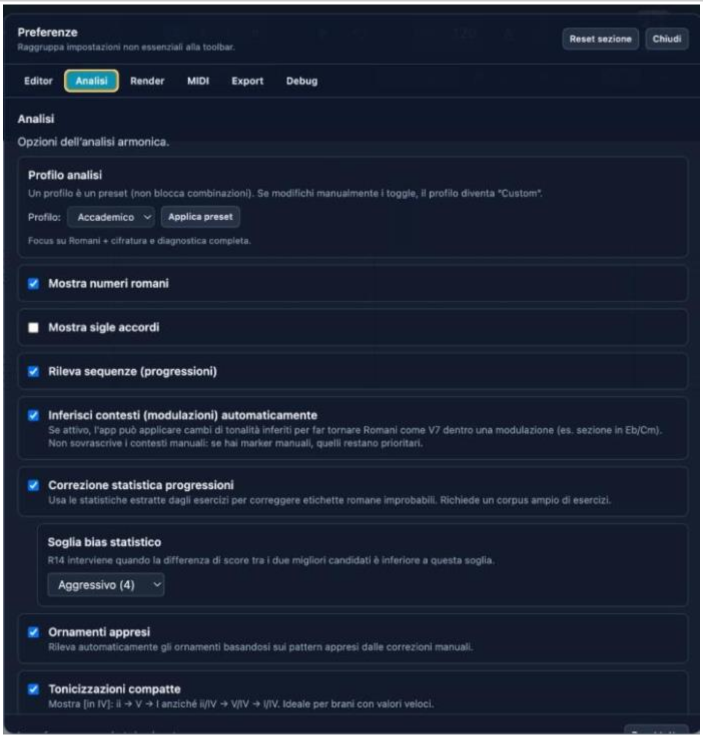


Figure 19 — Preferences: Analysis tab

- **Analysis profile:** Academic preset (default). Customisable.
- **Show Roman numerals / chord symbols:** display toggles.
- **Detect sequences:** recognition of imitated harmonic sequences.
- **Infer contexts:** automatic modulations (manual markers stay priority).
- **Statistical correction:** uses the corpus to correct unlikely labels.
- **Statistical-bias threshold:** Conservative, Moderate, Aggressive.
- **Learned ornaments:** uses patterns from manual corrections.
- **Compact tonicisations / Cadential patterns:** compact format and automatic cadence recognition.

17.3 Render / MIDI / Export / Debug tabs

Render: graphic rendering settings. MIDI: input/output devices. Export: format options (including MIDI format 0 and 1). Debug: advanced diagnostics.

> ***Note:*** *The “Reset all” button restores all settings. “Reset section” only the current tab.*

18. Keyboard shortcuts

18.1 Global

Shortcut	Action
Cmd/Ctrl + N	New project
Cmd/Ctrl + O	Open
Cmd/Ctrl + S	Save
Cmd/Ctrl + Shift + S	Save as
Cmd/Ctrl + I	Import MIDI
Cmd/Ctrl + Shift + E	Export MIDI
Cmd/Ctrl + Z	Undo
Cmd/Ctrl + Shift + Z	Redo
Cmd/Ctrl + P	Print

18.2 Editor

Shortcut	Action
Space	Play / Stop (starts recording if armed, with count-in)
Enter	Return to start
V	Cycle voice
1–7	Duration
# / b / n	Sharp / Flat / Natural
.	Augmentation dot
R	Note / Rest (on ACC tracks)
Shift + R	Arm / disarm / stop recording (with a visible ACC track)
L	Tie
K	Metronome
T	Measure-edit panel
Option/Alt + F	Fermata
Opt + H	Mark the selected note(s) as structural (enters the analysis)
Opt + Shift + H	Collapse the selection (≥2 notes, even an arpeggio) into a single chord
Shift + Opt/Alt + R	Tempo curve (rallentando/accelerando), also on ACC selections and loaded files
Option/Alt + ↑/↓	Move note to treble/bass staff
Option M	Insert close/open score from the playhead onward
Alt + L	Cycle layout
Alt + H	Toggle labels
Cmd/Ctrl + / –	Zoom in/out
Cmd/Ctrl + 0	Reset zoom
Backspace	Delete selection
↑/↓	±1 semitone
Alt + ↑/↓	±1 octave
Shift + Cmd/Ctrl + C	Copy the entire selected SATB voice (then paste onto an ACC track)

> ***Note:*** *On different keyboard layouts, the # key may arrive as à, and the . key may produce > with Shift: the app accepts both.*

18.3 Ornament marking (Option/Alt + letter)

Shortcut	Ornament
Option+H	Structural
Option+P	Passing tone
Option+V	Neighbor tone
Option+A	Appoggiatura
Option+N	Anticipation
Option+S	Escape tone
Option+R	Suspension/Retardation
Option+O	Ornamental
Option+C	Changing tone

18.4 Analysis shown and staves (View menu)

Shortcut	Action
Cmd/Ctrl + Alt + R	Roman numerals
Cmd/Ctrl + Alt + S	Chord symbols
Cmd/Ctrl + Alt + F	Figured bass
Cmd/Ctrl + Alt + C	Choir (SATB): show or hide
Cmd/Ctrl + Alt + 3	Violations list

Accompaniment tracks are switched on and off from View → “Staves shown and exported”, each by its own name. These commands also choose the contents of the exported files (§13.2).

19. Melodic transformations

Harmony Tutor provides tools to study and compose with the transformations of a melodic motif: **transposition** (T), **inversion** (I), **retrograde** (R) and **retrograde-inversion** (RI) — the symmetries at the heart of imitation and thematic development. Transformations can be **tonal** (within the key) or **real** (at exact intervals).

19.1 Motif detector

The detector automatically finds, throughout the piece, occurrences of the same motif transformed — transpositions, inversions, retrogrades and retrograde-inversions — both **within a single voice** and **across voices** (cross-voice), in tonal or real form.
Activation. The “Melodic motifs” toggle. When active, the notes belonging to a recognised transformation are **colour-coded by type**, and a **bracket** marks the reference model motif.
Note. In very dense textures the matches may be hard to see; that is why the detector is off by default.

19.2 Transformation tools

They let you **generate** a new phrase by transforming a selection of notes:

- **Transposition** — moves the selection by a chosen interval (tonal or real).
- **Inversion** — flips the melodic contour around an axis (per voice).
- **Retrograde** — reverses the temporal order of the notes (onset-based).
- **Retrograde-inversion** — combines inversion and retrograde.

Tonal transformations stay within the key; **real** ones keep the exact intervals (a warning flags when a real inversion leaves the scale). The result is **inserted after** the selection, extending the measures automatically if needed.

19.3 View and layout

Under More ‣ View you can choose between the page, where lines wrap, and the continuous ribbon, which puts every measure on a single staff that scrolls horizontally. The layout follows the zoom: zooming out fits more measures per line instead of leaving empty space on the right.

Content-aware spacing, in the same menu, gives each measure the width its content needs: a measure of eighth notes takes more room than one of whole notes, instead of cramming them. Analysis labels follow the layout in every case.

System breaks fixed by hand

Measures per line is a good ceiling to start from, but it does not know where a phrase ends. ⌥Enter ends the line at the measure where the caret is; the same is available from the right-click menu on a note (“break the line after this measure”).
On very dense music the line may close before the number you set: if the notes would not fit without crowding, the program wraps early. The number is a ceiling, not an obligation.
A break fixed by hand overrules everything else: neither the measures-per-line ceiling nor the available width can move it. You can see it — a ⏎ mark at the end of the fixed line — and you click it to remove it. Without a visible mark, layout would become a list of invisible decisions with no way back.

Measures per line, view and paper format stay with the file

The three layout settings — the measures-per-line ceiling, the on-screen view (systems or continuous ribbon) and the paper format (A4 portrait or landscape, which concerns printing and PDF) — are now saved in the project and restored when you reopen it. They are choices about the PIECE, not about the working session: a close-spaced chorale and a keyboard piece are not read on the same page.
Files saved with earlier versions do not carry them: they open with the starting values, without inheriting the layout of the file opened before.

Back to 100%: double-click on empty space

Zoom resets with a double-click on an empty part of the page — where nothing is drawn. Anything under the pointer (a note, a tie, a sign, a label, the staff lines, the clef) blocks it. A single click used to be enough, and the zoom could vanish just because you rested the pointer on the page while working. ⌘0 and the toolbar command are still there.

20. Score signs

Signs are placed from the signs palette, opened with the *f* button in the toolbar. It is a floating window: drag it by its title, and it stays open while you work.

20.1 Using the palette

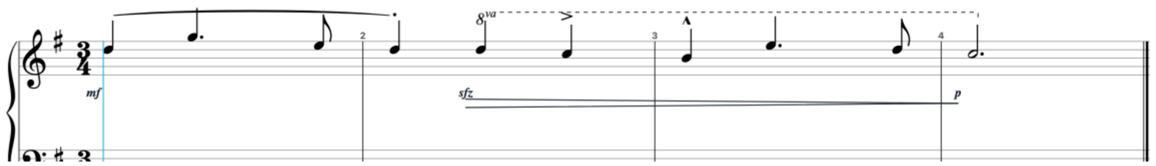
Signs are collected in three collapsible groups, by what the sign is about: Dynamics (intensity), Articulations and expression (how sounds are attacked and joined), Structure (the layout of the page). One group opens at a time, and stays open while you drag.

The palette docked to the left edge: the search box at the top, and several groups open at once.

Drag: grab the sign and drop it where you need it on the score.
Or: select a note and click the sign. A hairpin or a slur needs two notes.
Right-click a sign to remove it, wherever it is: on the note, on the curve, on the text.
From 1.7 the palette is docked to the left edge and its body scrolls: several groups can stay open at once, and a search box at the top takes you straight to the sign you need, instead of opening and closing sections by trial.
It now also holds durations, accompaniment patterns and melodic transformations; the “Notes” group gathers everything that concerns the note itself (accidentals, dot, tuplets, articulations, octave signs).
Transposition, mode and the automatic leading note sit next to the key signature; the time signature is written with its two numbers.
Signs are picked up and put down: hairpins are drawn by dragging them onto the music instead of being written as text, and accidentals are proper icons rather than text characters.
Besides dragging them, signs can be placed with a single click: pick the sign from the palette and it is applied where the cursor is.

20.2 Dynamics

Levels from ppp to fff, accents (sf, sfz, rf), fp, and crescendo/diminuendo hairpins. They are not drawings: they drive playback volume, and they apply to every voice — a forte is written once and concerns the whole texture, as in choral writing.



Some signs at work: dynamics, a hairpin, a slur, staccato, an accent and an 8va.

Hairpins are stretched by dragging their ends. On a held string note a hairpin swells the sound inside the note; a piano, once the key is struck, cannot grow.
With no signs at all, the notes keep the dynamics they came with (from a MIDI import or a recording); as soon as a sign is written, the sign takes over.

20.3 Articulations

Staccato, staccatissimo, accent, marcato, tenuto. They combine — accent and staccato on the same note is ordinary writing — and are placed on a note or on a whole chord.
Dragged onto a chord they apply to the whole chord, because the sign is drawn once above the group and could not say «the soprano only».
Selecting specific notes and clicking applies to exactly those: that is how you staccato the soprano alone.
They sound: staccato shortens the note and its tail, the accent reinforces the volume. Tenuto is a written sign and does not change the sound, because full length is already the norm.

20.4 Slurs

Not to be confused with the tie (key L), which joins two equal sounds into one. A slur connects different notes and is a sign of phrasing.
Place it by selecting two notes and clicking the sign, or by dragging it onto a note: it then reaches the next note of the same voice.
Lengthen or shorten it by dragging the ends of the curve, where a handle appears on hover.
If the passage breaks across a line, the curve stops at the edge of the staff and resumes on the next system.
In the choir the curve sits on the stem side — above for soprano and tenor, below for alto and bass — so one voice's slur does not land on the other.

20.5 Octave signs (8va and 8vb)

For passages that would otherwise need too many ledger lines: they are written an octave lower (or higher) and the sign says to play them shifted.
Notes are not moved: you write where it should be read, and the sign changes how it sounds.
To tidy a passage already written high: select it, move it down an octave with ⇧+down arrow, then add the sign.
The sign applies to the voice it is written in: an 8va on the soprano does not raise the bass sounding at the same time. It also covers the note under the end of the bracket.

20.6 What survives in files

All of these signs are written to MusicXML and read back when the file is reopened: dynamics, articulations, slurs, octave signs and key changes. MIDI carries dynamics (as note velocity) and octave signs (as sounding pitch), which is all that format can say.

20.7 Moving signs on the page

A metronome mark landing on a chord symbol, or two texts overlapping, can be moved out of the way by dragging: the same gesture carries the sign to another point in TIME and to another HEIGHT. Where you release decides the measure, the vertical shift decides the height, and the offset stays with the sign in the file as well.
Engraving has no rule for every possible collision between signs: whoever writes moves them just as much as is needed.
A free text (“dolce”, “CVII”) is corrected with a double-click on it, without deleting and retyping.

21. Frequently asked questions

What is the difference between the free version and Harmony Tutor Pro?
The free version includes all features for 10 days. After the trial period, the advanced features (harmonic analysis, chorale generator, export) are disabled. Buying the Pro licence unlocks everything with no time limit.

Can I install the software on a new computer?
Yes. Each licence allows 2 activations. If you change computer, you can deactivate the licence on the old device and reactivate it on the new one.

The analysis shows the wrong chord. What can I do?
Place the playhead at the desired point, press T and enter an override in the panel. Incomplete chords or tonicisations can create ambiguity.

What is CAD-DEC?
The deceptive cadence: V resolves to vi/VI/$\flat$VI instead of I. It is flagged with a green marker in the panel.

What does V^{6}_{4} mean instead of I^{6}_{4}?
In the cadential progression, the I^{6}_{4} is functionally relabelled as V^{6}_{4}. The override is available from the T panel.

How do I insert a rallentando?
Select the range of notes → Shift+Option/Alt+R → set the start and end BPM.

How does the Fermata work?
Select the note → Option/Alt+F. The duration is doubled in playback (default 2×). Repeat to remove it.

How do I add an ACC track?
Tracks panel → “+ New” → choose the type, then assign a name and a MIDI instrument.

Does REC recording work with ACC tracks?
Yes. Select the destination ACC track, arm with Shift+R and start with Space.

Can I import files from MuseScore, Finale or Sibelius?
Yes, via the MusicXML format. Export from those programs and in Harmony Tutor choose File → Import MusicXML. MIDI import is also supported.

What is the difference between MIDI format 0 and format 1?
Format 0: all voices on a single track. Format 1: each voice on a separate track (recommended for editing in a DAW).

Third-party credits and licences